

St. Joseph's College of Commerce

COST MANAGEMENT - I

B.Com – Regular Semester V

Module – 1 : PROCESS COSTING

Meaning of Process: Process is a distinct stage in manufacturing or production, wherein Raw Material is converted into one identifiable form into another, before it is finally converted into the saleable final product.

Meaning of Process Costing: Process Costing is a method of costing used to ascertain the cost of a product at each process or stage of manufacture. In this method, the costs of materials, wages and overheads are accumulated for each process separately, for a given period, and then carried forward cumulatively from one process to the next process till the last process is completed.

Definition of Process Costing:

CIMA London defines process costing as “that form of operation costing which applies where standardised goods are produced.”

The following is the list of industries where process costing is applied:

Chemicals	Milk dairies
Soap	Rubber
Paper	Steel
Biscuits	Cement
Oil Refining	Sugar
Textiles, weaving, spinning	Mines
Food products	Meat Products
Coke	Glass etc.

General Principles of Process Costing:

1. The majority of items of cost can ordinarily be identified with specific processes and collected and accumulated separately for each period.
2. Production records of each process are so designed as to show the quantum of production for each period.
3. The total cost of each process is divided by the total production by the process for arriving at the unit cost of the article processed.
4. The cost of any normal spoilage or wastage is included in the cost of the total units produced. As a result, the average cost per unit increases.
5. As the product travels from one process to another, the cumulative

cost thereof in respect of the processes it has already undergone is transferred to the account of the process it has yet to undergo.

Determination of Unit Cost: To determine the unit cost of output of each process, the total production cost of the process is divided by the total quantity of the output of the process during a given period.

Features of process industries:

1. Production is a continuous process, except during shut-down for repairs, etc.
2. The finished product of one process becomes the raw materials for the next process.
3. Products and process are completely standardized.
4. Different products with or without by-products are simultaneously produced at one or more processes.
5. Output is uniform and is usually manufactured for stock and not for the specific order.
6. Unavoidable 'normal' wastage usually arises at different stages of manufacture, for reasons like evaporation and chemical reaction, etc.
7. In certain industries, the by-products may require further processing before they can be sold.
8. A main product of one firm may be a by-product of another firm, which may be available in the market at prices which are lower than the cost of the first mentioned firm. It is therefore, essential, that the comparative costs are known so that advantage can be taken of market conditions.
9. In order to obtain accurate average cost, it is necessary to measure the production at various stages of manufacture. For this, the work-in-progress in each process, if any, is converted into equivalent units of finished products.
10. Separate Account is prepared for each process to ascertain cost thereof.

Processes having losses:

In many industries, some loss is inevitable. The process loss may be in the form of loss in weight, scrap and or waste. Process loss may be normal or abnormal.

Normal Process Loss: Normal Process Loss is unavoidable on account of the inherent nature of material. Such losses can be estimated in advance on the basis of past experience or technical specifications. In many cases, when the scrap or waste fetches no value, normal process loss is recorded only in terms of quantity and the cost per unit of good production is increased accordingly. If the scrap/waste realizes some value, the value is credited to the process account to arrive at the normal cost of normal output.

Abnormal Process Loss: Any loss caused by unexpected or abnormal conditions such as sub-standard material, carelessness, accident, or losses in excess of the margin anticipated for normal process loss is regarded as abnormal process loss.

Abnormal Loss = Total Process Loss - Normal Loss

Abnormal loss arises only when Process Loss > Normal Loss.

The value of such losses is calculated as follows:

$$\text{Value of Abnormal Loss} = (\text{Normal cost of normal output} / \text{Normal Output}) * \text{Units of abnormal loss}$$

Abnormal Gain: Abnormal Gain is the unexpected gain in production under normal conditions. If the actual process loss is less than the estimated normal loss, the difference is considered as Abnormal Gain. The concerned process account is debited with the quantity and value of abnormal gain and the abnormal gain account is credited. The abnormal gain account is debited with the difference in the quantity and value between normal loss and actual loss and the actual under recovery of normal scrap, respectively. The balance of the abnormal gain account is transferred to the costing profit and loss account.

Abnormal Gain = Actual production - Expected Production or Abnormal Gain = Normal Loss - Process Loss
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Abnormal Gain arises only when Normal Loss > Process Loss.

Equivalent Production: Equivalent production means converting the incomplete production into its equivalent completed units. In each process, an estimate is made of the percentage completion of work-in-progress with regard to different elements of costs, viz., material, labour and overheads. The formula for computing equivalent completed units is:

$$\text{Equivalent completed unit} = \text{Actual no. of units in process of manufacture} * \% \text{ of work completed.}$$

Example: Closing Work in Progress consists of 2500 units each 20% complete. Hence Equivalent Units = 2500 * 20% = 500 units. Hence cost of closing WIP is equal to cost of 500 fully completed units.

Inter-process Profits: When the output from one process is transferred to the next process at the cost of production plus a pre-determined margin of profit or at market price, there arises inter-process profits.

When inter-process transfers are made at profit, the values of process stocks and finished stock include unrealized profits. These values thereafter require adjustments to the extent of the unrealized profit to show them in the balance

sheet. The amount of unrealized profit included in the closing stock of each process can be calculated as follows:

Unrealised Profit = Closing stock * (Transfer price / Total cost of transferee process) * % of profit on transfer price added in the transferor process.

Advantages:

- a) Each process is made to stand by itself as to its profitability.
- b) It is possible to compare cost of output and market value at each stage of production.
- c) It is easier to fix responsibility of process managers for cost control, through indirect means of achieving higher profits.

Disadvantages:

- a) It is a complicated method involving transfer-pricing considerations.
- b) It might promote conflicts and misunderstanding among managers on comparative profitability.
- c) It requires reconciliation between profits booked at each stage and actual realized profit, since it shows profits locked up in unsold stock.

Joint Products and By-products: When from a common manufacturing operation or process of a single input, two or more outputs are obtained, depending upon the relative importance of the products, they are referred to as joint products or by-products. Such products are common in chemical, extractive dairy and agricultural product industries. For example, in oil refineries when crude oil is processed, petrol, diesel, kerosene, lubricant, etc., are obtained. Similarly in sugar manufacturing industries molasses is also produced along with sugar.

Joint Products: Two or more products produced from the same process or operation, considered to be of relatively equal importance. For example, in oil refineries when crude oil is processed, petrol, diesel, kerosene, lubricant, etc., are obtained. They are Joint Products.

By-products: By-products refers to incidental waste, arising during the course of manufacture, which has a commercial name and value. It refers to secondary or subsidiary products arising during the manufacturing process. For example, Molasses arising in the manufacture of sugar, Glycerine obtained in the manufacture of soap, etc.

Differences between Joint Products and By-products:

Basis	Joint Products	By-Products
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Meaning	Two or more products, separated in the course of the same processing operation, considered as relatively equally important.	Products recovered from material discarded in a main process.
Nature	Intentionally manufactured	Incidentally arises during the process.
Importance	Comparatively higher sale value	Comparatively lower sale value

Differences between Job Costing and Process Costing:

Job costing	Process costing
Goods are manufactured only against specific orders.	Production is of like units in continuous flow.
Costs are accumulated and applied to specific jobs.	Costs are accumulated and applied process-wise or department-wise.
Costs are computed after every job is completed.	Costs are computed after the expiry of a particular cost period.
Different jobs are independent of each other.	Production being in a continuous flow, products are intermingled in such a manner that lots are not distinguishable.
Products are normally not transferred from one job to another except in the case of surplus work or excess production.	Costs are normally transferred from one process to another. Generally the finished product of the process becomes the raw material of the next process until the goods are completely manufactured.
From the point of view of managerial control, more attention is needed because production is not in continuous flow and each job is different.	Because of the standard, mass and continuous production, managerial control is easier.
Different jobs may or may not have opening or closing work-in-progress.	As the production is in continuous flow there is always an opening and closing balance of work-in-progress.

Process Account Specimen

<i>Particulars</i>	<i>Units</i>	<i>Amount in</i>	<i>Particulars</i>	<i>Units</i>	<i>Amount in</i>
		₹			₹
To Basic material	XX	XXX	By Normal Loss	XX	XXX
To Direct Material	XX	XXX	By Abnormal Loss	XX	XXX
To Direct labour		XXX	By Process II (Output transferred to next processes)	XX	XXX
To Direct Expenses		XXX	By Finished stock Account	XX	XXX
To Production overhead		XXX			
To Cost of rectification of defective material		XXX			
To Abnormal gains	XX	XXX			
	XX	XXX		XX	XXX

PROBLEMS

1. There are three distinct processes A, B, and C. During the four-week period 1,000 units are produced and the following information is available.

Particulars	Process-A	Process-B	Process-C
Direct materials	2,000	1,000	----
Direct wages	1,500	700	800
Direct expenses	300	100	----

The indirect production cost were Rs. 4,500 and these are to be apportioned on the basis of direct wage cost. Prepare the process accounts.

2. The following details show the cost of the three processes of manufacturing. The production of each process is passed on to the next till completion.

Particulars	Process - A	Process - B	Process - C
Wages and materials	60,800	24,000	58,500
Works overheads	11,200	10,500	12,000
Production in units	72,000 units	75,000 units	96,000 units
Opening stock (units from preceding process on 1 st July)		8,000 units	33,000 units
Closing stock (units from preceding process on 31 st July)		2,000 units	11,000 units

3. From the following information relating to an article which undergoes three processes for manufacture, show the cost of each process and the cost per article.

Particulars	Process-A	Process-B	Process-C
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250

The indirect expenses amounted to Rs. 21,250 and the number of articles produced during the month was 240 units.

4. A particular brand of chemical passed through three processes during the week ended 15th January. 600 tins were produced. The following information is disclosed by each process.

Particulars	Process-A	Process-B	Process-C
Materials	4,000	2,000	1,500
Labour expenses	3,000	2,500	2,300
Direct expenses	6,000	200	500
Cost of tins		2,030	

Packing charges			325
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The indirect expenses were Rs. 1,600. The by product is sold for Rs. 240 (process B). The residue was sold for Rs. 125.50 (Process C).

Prepare the process A/c with cost of production of the product per tin.

Normal & Abnormal Loss and Abnormal Gain

5. Prepare a Process Account and Abnormal Loss Account from the following information.

Input of Raw material	1000 units @ ₹ 20 per Unit
Direct Material	₹ 4,200/-
Direct Wages	₹ 6,000/-
Production Overheads	₹ 6,000/-
Actual output transferred to process II	900 units
Normal Loss	5%
Value of Scrap per unit	₹ 8/-

6. The product of a company passes through 3 distinct process. The following information is obtained from the accounts for the month ending January 31, 2024.

Particulars	Process – A	Process – B	Process – C
Direct Material	7800	5940	8886
Direct Wages	6000	9000	12000
Production Overheads	6000	9000	12000

3000 units @ ₹ 3 each were introduced to process – I. There was no stock of materials or work in progress. The output of each process passes directly to the next process and finally to finished stock A/c.

The following additional data is obtained:

Process	Output	Normal Loss in %	Realisable Value of Scrap
Process 1	2,850	5%	2
Process 2	2,520	10%	4
Process 3	2,250	15%	5

Prepare Process Cost Account, Normal Loss Account and Abnormal Gain or Loss Account

7. Product A is obtained after it passes through three distinct processes. 2,000 kgs of material at Rs. 5 per kg were issued to Process 1, direct wages amounted to Rs. 900 and other production overheads incurred was Rs. 500. Normal loss is estimated at 10% of input. The wastage is sold at Rs. 3 per kg. The actual output is 1,850 kgs. Prepare Process 1 A/c, Normal loss A/c and Abnormal Gain A/c.

8. A product passes through 2 distinct processes A and B. The normal wastage of each process is as follows:

Process A – 3% of products entering the process.

Process B – 5% of products entering the process.

The wastage of process A was sold at 50 paise per unit and that of process B at Re. 1 per unit.

10,000 units were issued to process A at Rs. 2 per unit.

The other expenses were as follows:

	Process A	Process B
Materiala	Rs. 2,000	Rs.3,000
Wages	Rs. 10,000	Rs. 16,000
Overheads	Rs. 2,100	Rs. 2,375
Actual output (units)	9,500	9,100

Prepare process accounts and other ledger accounts.

9. The following details are extracted from the costing records of an oil mill for the year ended 31st March, 2015.

Purchase of 5,400 tonnes of coconut of Rs. 2,20,000.

	Crushing	Refining	Finishing
Cost of Labour	2,750	1,100	1,650
Electirc Power	660	396	264
Sundry Material	110	2,200	---
Repairs to Machinery	308	363	154
Steam	600	495	495
Factory Expenses	1,452	726	242

Cost of Sacks – Rs. 8,250

3,200 tonnes of crude oil were produced. 2,600 tonnes of oil produced by the refining process. 2,550 tonnes of refined oil were finished for delivery.

Coconut sacks sold	Rs. 440
1,925 tonnes of coconut residue sold	Rs. 12,100
Loss in weight in crushing	275 tonnes
500 tonnes of by-products obtained from refining process	Rs. 7,425

You are required to show the accounts in respect of each of the following stages of manufacture for the purpose of arriving at the cost per tonne of each process and the total cost per tonne of the finished oil.

(a) Coconut crushing process. (b) Refining process. (c) Finishing process.

Joint and By-product

10. In manufacturing the main product a company processes the incidental waste into two by-products A and B. From the following data relating to the products prepare a comparative profit & loss statement showing individual cost and other details. The total cost up to separation point was Rs. 3,10,400

Additional information:

	Main product	Product A	Product B
Sales	8,00,000	64,000	96,000
Cost after separation	80,000	12,800	14,400
Estimated net profit (% of sales)	?	20%	30%
Estimated selling expenses (% in sales)	20%	10%	15%

11. A factory producing an article 'X' also produces product 'Y' which is further processed into a finished product. The joint cost of manufacturing is given below:

Materials	50,000
Labour	30,000
Overheads	20,000
	1,00,000

The subsequent costs are as follows:

	X	Y
Materials	30,000	15,000
Labour	14,000	10,000
Overheads	<u>6,000</u>	<u>5,000</u>
	50,000	30,000
Selling price	1,60,000	80,000
Estimated profit on selling price	25%	20%

Assume that the selling and distribution expenses are in proportion to the selling price. Show how you would apportion the joint cost of manufacturing and prepare a statement showing the cost of production or both products and the ledger accounts.

12. In a manufacturing concern a certain product is manufactures which also yields 2 by-products. The joint expenses of manufacture are:

Materials	8,500
Labour	9,000
Overheads	7,500
	25,000

Subsequent expenses are as follows:

	A	B	C
Materials	2,500	1,200	1,400
Labour	1,900	1,600	2,000
Overheads	1,500	900	1,050
	5,900	3,700	4,450

The selling prices are A - 30,000; B - 20,000 and C - 15,000. Estimated profits on selling prices are A - 40%; B - 30% and C - 25%.

Show the apportionment of joint cost and prepare the product accounts.

13. Bright Chemical Ltd. Electrolyses common salt to obtain 3 joint products - caustic soda, chlorine and hydrogen. During a costing period, the expenditure relating to the inputs of the common process amounted to RS. 3,50,000. After separation, expenses amounting to Rs. 1,60,000, Rs. 75,000 and Rs. 10,000 were incurred for caustic soda, chlorine and hydrogen respectively. The entire production was sold and Rs. 3,75,000, RS. 2,50,000 and Rs. 60,000 were realised for caustic soda, chlorine and hydrogen respectively. The selling expenses were estimated at 5% of realisation from sale. The profit is 15%, 10% and 5% of realisation of realisation from sale of caustic soda, chlorine and hydrogen respectively.

Draw a columnar statement showing the apportionment of joint cost and the profitability of each product.

14. A by-product 'Beta' is derived is derived in the course of manufacturing product of Alpha. The by-product is further processed for sale. From the following data available from the records, prepare an account showing the cost per kg of the product 'Alpha' and the by-product 'Beta'.

	Joint Expenses	Separate Expenses	
		Alpha	Beta
Materials	10,000	6,000	500
Labour	7,000	5,000	2,000
Overheads	2,500	1,500	600

The quantities produced during the period under consideration were: Alpha - 1,000 kgs and Beta - 500 kgs. The selling price was Rs. 120/ kg on which the project earned was 30%.

Inter Process Profit

15. A product passes through two processes X and Y. The output of Process X is charged to process Y at a price which includes profit of 20% on actual cost and the output of Process Y is charged to finished stock A/c at a price which

includes 10% profit on actual cost. The following data is provided for the month of July.

	X	Y
Material (2,500 units)	1,250	----
Labour	625	500
Overheads	1,875	750
Indirect Material	---	1,250

There was no partly finished WIP. Out of the finished stock, 1,500 units had been sold for Rs. 7,500. Prepare the Process A/c and Finished Stock A/c.

16. A product passes through three processes to completion. These processes are known as A, B and C. The output of each process is charged to the next process at a price calculated to give a profit of 20% on the transfer price. The output of Process C is charged to finished stock on a similar basis.

There was no partly finished WIP in any process on December 31st, on which day the following information was obtained.

	Process A	Process B	Process C
Materials	4,000	6,000	2,000
Labour	6,000	4,000	8,000
Stock: 31 st Dec	2,000	4,000	6,000

There was no stock in hand on Jan 1st and overheads were ignored. Of the goods passed into finished stock, Rs. 4,000 remained in hand on Dec. 31st and the balance has been sold for Rs. 36,000. Show the Process A/c and calculate reserve for unrealised profits.

Equivalent Production

When there is no opening work-in-progress and no process losses:

17. In Process A on March 2st, there was no opening work-in-progress. During the month of March, 2,000 units of material was issued at a cost of Rs. 18,000. Labour and overheads totalled to Rs. 9,000 and Rs. 6,600 respectively. On 31st March 1,500 units were completed and transferred to the next process. Of the remaining 500 units which are incomplete, degree of completion was as follows:

Material 100%, Labour 60% and Overheads 30%

Prepare a. Statement of equivalent production.

b. Statement of cost

c. Statement of evaluation

d. Process account.

18. During January 2,000 units were introduced into process 1. The normal loss was estimated at 5% on input. At the end of the month 1,400 units had been produced and transferred to the next process. 460 units were incomplete and 140 units had been scrapped. It was estimated that incomplete units had reached a stage in production as follows: Material 75% complete. Labour 50% complete and overheads 50% complete.

The cost of 2,000 units introduced was Rs. 5,800. Direct material introduced during the process amounted to Rs. 1,440. Direct labour cost was Rs. 3,340. Production overheads incurred were Rs. 1,670. Units scrapped realised Re. 1 each. The units scrapped have passed through the processes so were 100% complete as regards to material, labour and overheads.

You are required to:

- Prepare a statement of equivalent production.
- Evaluate the cost of abnormal loss, finished goods and closing stock and
- Prepare the process 1 a/c and abnormal loss a/c.

19. Prepare statement of equivalent production statement of cost and process a/c from the following information.

Units introduced 7,600

Output (units) 6,000

Process cost

Materials Rs. 14,560

Labour Rs. 21,360

Overheads Rs. 14,240

Degree of completion for closing work-in-process

Materials 80%

Labour 70%

Overheads 70%

When there are process losses without opening work-in-progress

20. Prepare the process account from the following

Number of units introduced into the process – 4,000 units

Units completed and transferred – 3,000 units

Work-in-progress at the end of the period – 800 units (stage of completion material 80%, labour 70% and overheads 70%)

Normal loss at the end of the period estimated 200 units (scrap value at Re. 1/unit)

Value of Materials Rs. 7,480, Wages Rs. 10,680 and Overheads Rs. 7,120.

When there is opening work-in-progress

21. A company follows process costing and manufactures a product in one process. The work-in-process at the end of each month is valued according to FIFO method. At the beginning of the month of January, the inventory of WIP showed 400 units. 40% completed value as follows:

Materials	Rs. 3,600
Labour	Rs. 3,400
Overheads	<u>Rs. 1,000</u>
	<u>Rs. 8,000</u>

In the month of January, materials were purchased for Rs. 75,000. Wages and overheads in the month amounted to Rs. 79,800 and Rs. 21,280 respectively. Actual issue of materials to production was Rs. 68,500. Finished production taken into the stock in the month was 2,500 units. There was no loss in the process.

At the end of the month, the WIP inventory was 500 units and 80% complete as regards materials and 60% complete as regards labour and overhead.

You are required to compute equivalent production and prepare process account.

22. Prepare process 2 account from the following

Opening stock - 600 units at Rs. 1,050 (degree of completion Material 80%, Labour 60%, Overheads 60%)

Transfer from process 1 - 11,000 units at Rs. 5,500

Transfer to process 3 - 8,800 units

Direct material added in process 2 - Rs. 2,410

Direct labour Rs. 7,155, Production overhead Rs. 9,540

Units scrapped - 1,200 (degree of completion material 100%, labour 70% and overheads 70%)

Closing stock 1,600 units (degree of completion material 70%, labour 60% and overheads 60%)

There was a normal loss in the process at 10% of production. Units scrapped would realise 50 paise per unit.

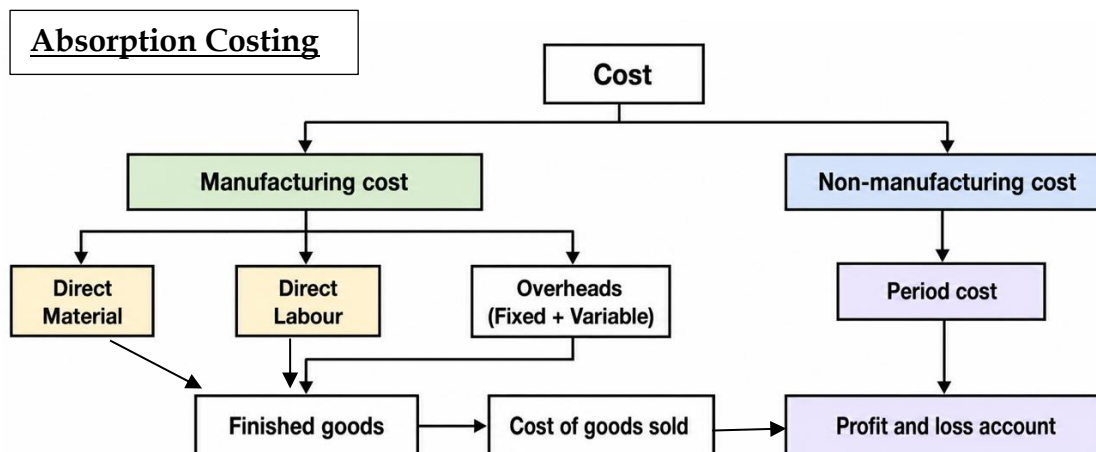
23. The following figures related to a single industrial process:

Opening stock (units)	10,000
Material	Rs. 2,250
Wages	Rs. 650
Overheads	<u>Rs. 400</u>
	<u>Rs. 3,300</u>
Units introduced (units)	40,000
Material	Rs. 9,250

MODULE - 2: MARGINAL COSTING AND ABSORPTION COSTING

ABSORPTION COSTING: Absorption Costing is a procedure of cost recognition wherein costs are classified based on functions. All costs of production, both fixed and variable, are included in inventory valuation. The presentation of profits under the Traditional Absorption Costing system is as under:

Sales	xxxx
Less:	
Direct Materials	xxxx
Direct Labour	xxxx
Factory Overheads	xxxx
Gross Profit/Loss	xxxx
Less:	
Administrative Expenses	xxxx
Selling and Distribution Expenses	xxxx
Net Profit/Loss	xxxx



Advantages of Absorption Costing:

1. It helps to conform to accrual and matching concepts which require matching cost with revenue for a particular period.
2. It has been recognized by various accounting bodies for preparing external reports and for valuation of inventories.
3. It avoids the separation of costs into fixed and variable elements which cannot be done easily and accurately.
4. It helps to make the managers more responsible for the costs and services provided to their centers/departments.
5. It helps to calculate the Gross Profit and Net Profit separately in the Income Statement.

Disadvantages of Absorption Costing:

1. Does not help in taking managerial decisions.
2. It does not help in cost comparison and cost control. An increase in the volume of output normally results in reduced cost per unit and vice-versa. This makes cost comparison and cost control difficult.
3. Many accountants argue that fixed overheads are period costs and do not produce future benefits, and therefore, should not be included in the cost of production.
4. It does not help in the preparation of Flexible Budgets.
5. It apportions the overhead costs on arbitrary methods, which ultimately lead to product cost inaccurate and unreliable.

MARGINAL COSTING: Marginal Costing is a technique of decision-making, which involves:

- a) Ascertainment of total costs,
- b) Classification of costs into Fixed cost and Variable Cost, and
- c) Use of such information for analysis and decision-making.

Marginal Costing is the ascertainment of Marginal Cost, and of the effect on Profit of changes in volume or type of output, by differentiating between fixed and variable costs.

Income Statement under Marginal Costing:

Sales	xxxx
Less: Variable Cost	xxxx
Contribution	xxxx
Less: Fixed Cost	xxxx
Net Profit/Loss	xxxx

Distinction between Variable Cost and Fixed Cost:

Particulars	Variable Cost	Fixed Cost
Meaning	It is that portion of cost, which changes or varies proportionately based on output / volume / quantity.	It remains constant for a given period of time, irrespective of the level of output during that period.
Items	Variable Cost = Direct Materials + Direct Labour + Variable Production Overheads + Variable Selling and Distribution Overheads.	Fixed Cost = Fixed Production overheads + Fixed Administrative overheads + Fixed Selling and Distribution overheads.

Examples	Raw Materials, Labour, Power, Royalty on production, etc.	Salary, Rent, Insurance, etc.
Cost per unit	Variable cost per unit is assumed to remain constant at all levels of output.	Fixed cost per unit will vary inversely with changes in the levels of output. As output increases, fixed cost per unit decreases, and vice-versa.
Point of incurrence	It is incurred only when production takes place.	Fixed Cost is incurred even at zero level of production.
Nature	Variable costs are considered as product-related costs.	Fixed costs are treated as period-related costs.
Inclusion in Inventory	In Marginal Costing, Variable costs are included in inventory valuation.	In Marginal Costing, Fixed costs are excluded from inventory valuation.

Semi-variable Expenses: These are expenses that exhibit characteristics of fixed and variable costs. Examples of semi-variables are delivery van expenses, telephone charges, repairs and maintenance, etc. By a systematic analysis, all semi-variable expenses can be segregated into variable and fixed portions.

Contribution: It is the excess of Sales Revenue over Variable Cost, i.e., Contribution = Sales less Variable Costs. It is the sum of Fixed Costs and Profit.

Profit Volume Ratio (PV Ratio) or Contribution to Sales Ratio: It is the relationship between Contribution and Sales Value. It is calculated as follows:

- a) $(\text{Total Contribution} / \text{Total Sales value}) * 100\%$
- b) $(\text{Change in Contribution} / \text{Change in Sales}) * 100\%$
- c) $100\% \text{ Less Variable Cost Ratio.}$
- d) $(\text{Contribution per unit} / \text{Sales per unit}) * 100\%$
- e) $(\text{Change in Profits} / \text{Change in sales}) * 100\%$

Advantages of Marginal Costing:

1. **Effective Cost Control:** It divides cost into fixed and variable. Fixed cost is excluded from product. As such, management can control marginal cost effectively.
2. **Treatment of overheads simplified:** It reduces the degree of over or under-recovery of overheads due to the separation of fixed overheads from production cost.
3. **Helpful to Management:** It enables the management in taking a number of decisions like Make or Buy, Discontinuance of a particular product, acceptance of export offers, etc.

4. **Break-even Analysis:** Marginal costing helps in break-even analysis, which shows the effect of increasing or decreasing production activity on the profitability of the company.
5. **Pricing Decisions:** Since Marginal Cost per unit is constant from period to period within a short span of time, firm decisions on pricing policy can be taken. If fixed cost is included, the unit cost will change from period to period depending upon the volume of output. This will make decision-making difficult.
6. **Helpful in budgetary control:** Classification of expenses into fixed and variable helps in budgetary control. It helps in the preparation of flexible budget for various levels of activities.

Disadvantages of Marginal Costing:

1. **Difficult to classify:** It is difficult to classify exactly the expenses into Fixed and Variable category. Most of the expenses are neither totally variable nor wholly fixed.
2. **Contribution is not final:** Contribution of a product itself is not a guide for optimum profitability unless it is linked with the Key Factor.
3. **Stock valuation:** Overheads of fixed nature cannot altogether be excluded particularly in large contracts, while valuing the WIP. This aspect is not considered in Marginal Costing.
4. **Naïve Assumption:** Some assumptions regarding the behavior of revenues and costs are not necessarily true in a realistic situation. For example, additional output can be sold only by reducing sale prices.
5. **Ignores time value:** Marginal Costing ignores time factor and investment. For example, the Marginal Cost of two jobs may be the same but the time taken for their completion and the cost of machines used may differ. The true cost of a job, which takes longer time and uses costlier machine, would be higher. The effect of time value of money is not analyzed by Marginal Costing.
6. **Claim for loss of stock:** Insurance claim for loss or damage of stock on the basis of such a valuation will be unfavourable to business.

Distinction between Marginal Costing and Absorption Costing:

Particulars	Marginal Costing	Absorption Costing
Cost Recognition	Only variable costs are included for product costing and inventory valuation.	Both variable and fixed costs are included for product costing and inventory valuation.
Classification	Classification is based on nature, i.e., fixed and variable.	Classification is based on function, i.e., production, administration and selling and distribution costs.

Treatment of fixed costs	Fixed costs are treated as Period Cost.	Fixed costs are charged to cost of production.
Presentation	Cost data presented highlight the Total Contribution and Contribution of each product.	Cost data are presented on conventional pattern. Net profit of each product is determined after subtracting fixed cost along with variable costs.
Effect if stockholding on cost per unit	Difference in the quantity of opening stock and closing stock does not affect the unit cost of production.	Difference in the quantity of opening and closing stock affects the unit cost of production due to the impact of related fixed cost.

Marginal Cost Equation: The Marginal Cost Statement when written in an equation form, constitutes the Marginal Cost Equation.

Sales - Variable Cost = Contribution

Contribution = Fixed Cost + Profit

Problems

1. The basic production data of ABC Ltd are as follows:

Normal volume of production = 19,500 units per period

Sale price - \$ 4 per unit

Variable cost - \$ 2 per unit

Fixed cost - \$ 1 per unit

Total fixed cost = \$19,500 (\$1 × 19,500 units, normal)

Selling and distribution costs have been omitted.

The opening and closing stocks consist of both finished goods and equivalent units of work-in-progress.

Present the profit and loss statement under Absorption and Marginal costing method.

2. From the following information, find out the amount of profit earned during the year using marginal cost technique:

Fixed Cost	Rs. 5,00,000
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Variable Cost	Rs. 10 per unit
Selling Price	Rs. 15 per unit
Output level	1,50,000 units

3. Find the profit from the following data:

Sales	Rs. 80,000
Marginal Cost	Rs. 60,000
Break-even point	Rs. 60,000

4. From the following particulars, calculate the break-even point:

Variable cost per unit	Rs. 12
Fixed Expenses	Rs. 60,000
Selling Price per unit	Rs. 18

5. From the following particulars, find out the selling price per unit if B.E.P. is to be brought down to 9,000 units.

Variable cost per unit	Rs. 75
Fixed Expenses	Rs. 2,70,000
Selling Price per unit	Rs. 100

6. From the following data, calculate break-even point expressed in terms of units and also the new B.E.P., if the selling price is reduced by 10%.

Fixed Expenses	Rs.
Depreciation	1,00,000
Salaries	1,00,000
Variable Expenses	
Materials	3 per unit
Labour	2 per unit
Selling Price	10 per unit

7. The sales turnover and profits during two periods are as under:

Period I : Sales Rs. 20 lakhs; Profit = Rs. 2 lakhs.

Period II: Sales Rs. 30 lakhs; Profit = Rs. 4 lakhs.

8. The following data are obtained from the records of a company:

	First Year (Rs.)	Second Year (Rs.)
Sales	80,000	90,000
Profit	10,000	14,000

Calculate the break-even point.

9. A company earned a profit of Rs. 30,000 during a particular year. If the marginal cost and selling price of a product are Rs. 8 and Rs. 10 per unit respectively, find out the amount of Margin of Safety.

10. From the following details find out:

- a) Profit Volume Ratio
- b) BEP
- c) Margin of Safety

	Rs.
Sales	1,00,000
Total cost	80,000
Fixed Cost	20,000
Net profit	20,000

11. The following information is obtained from a company for 2016:

	Rs.
Sales	20,000
Variable Cost	10,000
Fixed Cost	6,000

- a) Find P/V Ratio, Break-even point and Margin of safety at this level and
- b) The effect P/V Ratio, Break-even point and Margin of safety, if
 - i) 20% decrease in fixed cost
 - ii) 10% increase in fixed cost
 - iii) 10% decrease in variable cost
 - iv) 10% increase in variable cost
 - v) 10% increase in selling price together with an increase of fixed overheads by Rs. 1,200
 - vi) 10% decrease in selling price
 - vii) 10% decrease in sales price accompanied by 10% decrease in variable costs.

12. You are given the following data:

	Rs.
Variable Cost	6,00,000
Fixed Cost	3,00,000
Net Profit	1,00,000
Sales	10,00,000

Find:

- a) P/V Ratio
- b) B.E.P.
- c) Profit when sales amounted to Rs. 12,00,000
- d) Sales required to earn a profit of Rs. 2,00,000.

13. From the following find out:

- i) Profit Volume Ratio
- ii) Break-even point
- iii) Sales for 40% P/V Ratio
- iv) Margin of Safety, if actual sales were Rs. 3,00,000
- v) Net Profit, if actual sales were Rs. 3,00,000
- vi) Required Sales to earn a profit of Rs. 70,000
- vii) Required sales to earn a net profit of Rs. 70,000 after tax, given tax rate = 60%.

Position of the Co. for the year 2025

Sales	Rs. 2,00,000
Variable Cost	Rs. 1,50,000
Contribution	Rs. 50,000
Fixed Cost	Rs. 15,000
Profit	Rs. 35,000

14. Two manufacturing companies which have the following operating details decided to merge:

	<i>Company - I</i>	<i>Company - II</i>
Capacity utilization (%)	90	60
Sales (₹ In lakhs)	540	300
Variable costs (₹ In lakhs)	396	225
Fixed costs (₹ in lakhs)	80	50

Assuming that the proposal is implemented, calculate –

- Break-even sales of the merged plant and the capacity utilization at that stage.
- Profitability of the merged plant at 80% capacity utilization.
- Sales turnover of the merged plant to earn a profit of ₹ 75 lakh.
- When the merged plant is working at a capacity to earn a profit of ₹ 75 lakh, what percentage increase in selling price is required to sustain an increase of 5% in fixed overheads?

15. Rain Until September Co makes a product, the Splash, which has a variable production cost of \$6 per unit and a sales price of \$10 per unit. At the beginning of September 20X0, there were no opening inventories and production during the month was 20,000 units. Fixed costs for the month were \$45,000 (production, administration, sales and distribution). There were no variable marketing costs.

Required

Calculate the contribution and profit for September 20X0, using marginal costing principles, if sales were as follows.

(a) 10,000 Splashes (b) 15,000 Splashes (c) 20,000 Splashes

16. A company makes and sells a single product. At the beginning of period 1, there is no opening stock of the product, for which the variable production cost is `4 and the sale price is `6 per unit. Fixed costs are `2,000 per period of which `1,500 are fixed production costs.

The following details are available:

	<i>Period 1</i>	<i>Period 2</i>
Sales	1,200 units	1,800 units
Production	1,500 units	1,500 units

What would be the profit in each period using –

- Absorption costing. (Assume normal output is 1,500 units per period); and
- Marginal costing?

17. Mill Stream makes two products, the Mill and the Stream. Information relating to each of these products for April 20X1 is as follows.

Particulars	Mill	Stream
Opening inventory	Nil	Nil
Production (units)	15,000	6,000
Sales (units)	10,000	5,000
Sales price per unit	\$20	\$30
Unit costs	\$	\$
Direct materials	8	14
Direct labour	4	2
Variable production overhead	2	1
Variable sales overhead	2	3
Fixed costs for the month		
Production costs	\$ 40,000	
Administration costs	\$ 15,000	

Sales and distribution costs	\$ 25,000
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Required

- Using marginal costing principles, calculate the profit in April 20X1.
- Calculate the profit if sales had been 15,000 units of Mill and 6,000 units of Stream.

18. Big Woof Co manufactures a single product, the Bark, details of which are as follows.

	<i>Per unit \$</i>
Selling price	180.00
Direct materials	40.00
Direct labour	16.00
Variable overheads	10.00

Annual fixed production overheads are budgeted to be \$1.6 million and Big Woof expects to produce 1,280,000 units of the Bark each year. Overheads are absorbed on a per unit basis. Actual overheads are \$1.6 million for the year.

Budgeted fixed selling costs are \$320,000 per quarter.

Actual sales and production in units for the first quarter of 20X8 are given below.

	<i>January – March</i>
Sales	240,000
Production	280,000

There is no opening inventory at the beginning of January.

Prepare statements of profit or loss for the quarter, using

- Marginal costing
- Absorption costing

19. Eagle makes one product. Budgeted production and sales for the year just ended were 32,000 units. Actual sales for the year were 34,000 units at a selling price of \$34 per unit, and actual production was 36,000 units. Opening inventory was 5,000 units valued at the unit cost for the preceding year. Using an absorption costing system based on budgeted levels of output, unit costs for the year and the preceding year were as follows:

	<i>Year</i>	<i>Preceding year</i>
	\$	\$
Prime cost	24	22
Variable production overhead	3.5	1
Fixed production overhead	1.5	3
Selling and administrative overhead (fixed)	1	1
Total cost	30	27
Profit	4	3

Selling price	34	30
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Fixed production overhead for the year just ended was \$48,000 (budget and actual). Actual variable costs per unit were the same as budgeted.

Required:

Prepare statements of profit or loss for the year just ended on the basis that:

- (i) an absorption costing system is being used; and
- (ii) a marginal costing system is used.

20. Stoney is drafting a budget on the basis of the following data:

Direct material	\$10 per unit
Direct labour	\$5 per unit
Variable production expenses	\$8 per unit
Fixed production costs	\$27,000 per month
Normal output 9,000 units per month	90% capacity
Sales price	\$30 per unit

In order to build up inventory in anticipation of an increase in demand which is expected later in the year, production is to exceed sales in the first three months of the year as follows:

	<i>Month 1</i>	<i>Month 2</i>	<i>Month 3</i>
Production	6,500	9,000	10,000
Sales	5,000	8,500	9,500

Required:

Prepare two statements of profit or loss, each in comparative columnar form, covering

each of the three months:

- (i) on a marginal costing basis; and
- (ii) on a full absorption costing basis.

MODULE 3: COST-VOLUME-PROFIT ANALYSIS (CVP ANALYSIS)

CVP is the analysis of three variables, viz., cost, volume and profit, which explores the relationship existing amongst cost, revenue, activity levels and the resulting profit. It aims at measuring variations of profits and costs with volume, which is significant to business profit planning. It makes use of the Marginal Costing principles for planning and for making short-run decisions.

Break Even Point (BEP): It is the level of sales at which Total Contribution equals Total Fixed Cost. Hence, at that level, there is neither a profit nor loss to the firm. (Total Revenue = Total Costs, and Profit/ Loss = Zero)

- a) Break Even Point (Sales Value in Rs.) = Fixed Cost / PV Ratio.
- b) Break Even Point (Quantity) = Fixed Cost / Contribution per unit.

Margin of Safety (MOS): It represents the difference between the Actual Sales and Break-Even Point Sales. It can be expressed as a percentage of Total Sales, or in value, or in terms of quantity.

- a) Margin of Safety (in Rs.) = Total Sales minus BEP Sales or Profit / PV Ratio.
- b) Margin of Safety (in Quantity) = Total Sales Quantity – BEP Sales Quantity or Profit/ Contribution per unit.

Break Even Chart: It is the graphical representation of cost-volume-profit relationship. It depicts the following:

- a) Profitability of the firm at different levels of output.
- b) Break Even Point.
- c) Angle of Incidence (AOI): This is the angle at which the total sales line cuts the total cost line. If the angle is large, the firm is said to make profits at a high rate and vice-versa. A high AOI and high Margin of Safety indicate sound business conditions.
- d) Relationship between Variable Cost, Fixed Cost and Contribution.
- e) Margin of Safety representing the difference between Total Sales and Sales at BEP.

Profit Volume Chart/ Profit Volume Graph: It is a simplified form of break even chart and is an improvement over the break even chart as it clearly shows the relationship of profit to volume of sales.

The objectives of cost-volume profit analysis are given below:

- (1) In order to forecast profit accurately, it is essential to know the relationship between profits and costs on the one hand and volume on the other.

- (2) Cost-volume-profit analysis is useful in setting up flexible budgets which indicate costs at various levels of activity.
- (3) Cost-volume-profit analysis is of assistance in performance evaluation for the purposes of control. For reviewing profits achieved and cost incurred the effects on costs of changes in volume are required to be evaluated.
- (4) Pricing plays an important part in stabilizing and fixing up volume. Analysis of cost-volume-profit relationship may assist in formulating price policies to suit particular circumstances by projecting the effect which different price structures have on costs and profits.
- (5) As predetermined overhead rates are related to a selected volume of production, study of cost-volume relationship is necessary in order to know the amount of overhead costs which could be charged to product costs at various level of operation.

Uses of CVP Analysis

1. C.V.P. analysis helps in forecasting costs and profits as a result of change in volume.
2. It helps fixing a sales volume level to earn or cover a given revenue, return on capital employed, or rate of dividend.
3. It assists determination of effect of change in volume due to plant expansion or acceptance of an order, with or without increase in costs or in other words a quantum of profit to be obtained can be determined with change in volume of sales.
4. C.V.P. analysis helps in determining relative profitability of each product, line, project or profit plan.
5. Through cost volume-profit analysis inter-firm comparison of profitability can be done intelligently.
6. It helps in determining cash requirements at a desired volume of output, with the help of cash break-even charts.
7. Break-even analysis emphasises the importance of capacity utilisation for achieving economy.
8. From break-even analysis during severe recession, the comparative effects of a shut down or continued operation at a loss is indicated.

9. The effect on total cost of a change in the fixed over-head is more clearly demonstrated through break-even analysis and cost-volume-profit charts.
10. The conditions of a business such as profit potentialities, requirements of capital, financial stability and incidence of fixed and variable costs can be gauged from a study of the position of the break-even point and the angle of incidence in the break-even chart.

PROBLEMS

1. A manufacturer makes an average profit of Rs. 2.5 on a selling price of Rs. 14.5. He produces and sells 60,000 units at 60% capacity. His cost of sales per unit is:

Direct Materials	Rs. 4
Direct Wages	Rs. 1
Factory overheads (Variable)	Rs. 3
Selling overheads (Variable)	0.25 paise
Total fixed cost	Rs. 2,25,000

During the current year he anticipates that:

- (a) Fixed cost will increase by 10%
- (b) Material and labour cost will increase by 5% each.

Under these circumstances, he obtains an offer for further 20% utilization of his capacity. What minimum price would you recommend to accept the offer so as to ensure an overall profit of Rs. 1,60,000?

2. ABC Company produces a single product which is sold in the market at Rs. 75 per unit. The present production and sale is 40,000 units per month representing 50% of the capacity available. The cost data of the product are as follows:

Variable cost per unit Rs. 50

Fixed costs per month Rs.10 lakhs.

To improve the profitability the management has three proposals on hand as under:

- I. To accept an export supply order for 30,000 units per month at a reduced price of Rs. 60 per unit, incurring an additional variable cost of Rs. 5 per unit towards export packing duties, etc.
- II. To increase the domestic market sales by selling to a domestic chain stores 30,000 units at Rs. 55 per unit, retaining the existing sales at the existing price.
- III. To reduce the selling price for the increased domestic sale as advised by the sale department as under:

Reduce selling price per unit expected	Increase in sales
(inRs.)	(in units)
5	10,000
8	30,000
11	35,000

Prepare a table to present the results of the above proposals and give your comments and advice on the proposals.

3. A radio manufacturing company finds that while it costs Rs. 6.25 to make each component X 2730, the same is available in the market at Rs. 4.85 each, with an assurance of continued supply. The breakdown of cost is:

Materials	Rs. 2.75 each
Labour	Rs. 1.75 each
Other variables	Rs. 0.50 each
Depreciation and other fixed costs	<u>Rs. 1.25 each</u>
	Rs. 6.25 each

(a) Should you make or buy?

(b) What would be your decision, if the supplier offered the component at Rs. 5.85 each?

4. Indo-British Company has a capacity to produce 5,000 articles but actually produces only 2,000 articles for home market at the following costs:

	Rs.	
Materials		40,000
Wages		36,000
Factory overheads	- Fixed	12,000
	- Variable	20,000
Administration Overhead	- Fixed	18,000
Selling and Distribution Overheads	- Fixed	10,000
	- Variable	<u>16,000</u>
	Total Cost	<u>1,52,000</u>

The home market can consume only 2,000 articles at a selling price of Rs. 80 per article. An additional order for the supply of 3,000 articles is received from a foreign country at Rs. 65 per article. Should this order be accepted or not, if execution of this order entails an additional packing cost of Rs. 3,000?

5. A company currently operating at 80% capacity has the following particulars:

	Rs.
Sales	32,00,000
Direct materials	10,00,000
Direct labour	4,00,000
Variable overheads	2,00,000
Fixed overheads	13,00,000

An export order has been received that would utilize half the capacity of the factory. This order cannot be split, i.e., it has to be taken in full and executed at 10% below the normal domestic prices or rejected totally.

The alternatives are available to the management are:

- (1) Reject the order and continue with the domestic sales only (as at present),
or
 - (2) Accept the order, split capacity between overseas and domestic sales and turn away excess domestic demand or
 - (3) Increase capacity to accept the export order and maintain the present domestic sales by
 - (a) Buying an equipment that will increase the capacity. This will result in an increase of Rs. 1,00,000 in fixed cost and
 - (b) Increase the normal wage rate by $1\frac{1}{2}$ to increase their output.
- Prepare a comparative statement of profitability and suggest the best alternative.

6. Prestige Company Private Limited, manufacturing pressure cooker has drawn up the following budget for the year 20X5-X6.

Raw materials	20,00,000
Labour, stores, power, and other variable costs	6,00,000
Manufacturing overheads (Fixed)	7,00,000
Variable distribution costs	4,00,000
General overheads including selling (Fixed)	<u>3,00,000</u>
Total	40,00,000
Income from sales	<u>50,00,000</u>
Budgeted profit	<u>10,00,000</u>

The general manager suggests reducing selling price by 5% and expects to achieve an additional volume of 50%. There is sufficient manufacturing capacity. More intensive manufacturing program will involve additional costs of Rs. 50,000 for production planning. It will also be necessary to open an additional sales office at a cost of Rs. 1,00,000 per annum.

The sales manager, on the other hand, suggests increasing selling price by 10% which is estimated to reduce sales volume by 10%. At the same time, saving in manufacturing overheads and general overheads at Rs. 50,000 and Rs. 1,00,000 per annum respectively is expected on this reduced volume.

Which of these two proposals would you accept and why?

7. Present the following information to show to the management (a) the marginal product cost and the contribution per unit; (b) the total contribution and profits resulting from each of the following sales mixtures.

	Product	Per unit
Direct materials	A	Rs. 10
	B	Rs. 9
Direct wages	A	Rs. 3

B

Rs. 2

Fixed expenses Rs. 800

Variable expenses are allocated to products as 100% of direct wages.

Sales price A - Rs. 20

B - Rs.15

Sales mixtures:

- (i) 1,000 units of product A and 2,000 units of product B
- (ii) 1,500 units of product A and 1,500 units of product B
- (iii) 2,000 units of product A and 1,000 units of product B

8. A company manufactures three products. The budgeted quantity, selling prices and units are as under:

	A (Rs.)	B (Rs.)	C (Rs.)
Raw materials (@ Rs. 20 per kg)	80	40	20
Direct wages (@ Rs. 5 per hour)	5	15	10
Variable overheads	10	30	20
Fixed overheads	9	22	18
Budgeted production (in units)	6,400	3,200	2,400
Selling price per unit (in Rs.)	140	120	90

- (i) Present a statement of budgeted profit.
- (ii) Set optimal product mix and determine the profit, if the supply of raw materials is restricted to 18,400 kgs.

9. The following information in respect of Product A and Product B of a firm is given:

Particulars	Product A	Product B
Selling Price per unit	Rs. 75	Rs. 48
Direct Materials	Rs. 30	Rs. 30
Direct Labour hours (Re. 0.50 per hour)	15 hours	2 hours

Variable Overhead = 100% of direct wages.

Fixed Overheads = Rs. 3,000.

Present the above information to show the profitability of products during the labour shortage.

10. In a factory producing two different kinds of articles, the limiting factor is the availability of labour. From the following information from the factory for 2016, show which product is more profitable.

Particulars	Product A (Cost per unit) (Rs.)	Product B (Cost per unit) (Rs.)
Materials	5.00	5.00
Labour:		

6 hours @ 0.50	3.00	
3 hours @ 0.50		1.50
Overheads:		
Fixed 50% of labour	1.50	0.75
Variable	1.50	1.50
Total Cost	11.00	8.75
Selling Price	14.00	11.00
Total Production for the month	500 units	600 units

Also set the optimal product mix and the profitability, if the maximum labour hours per month is 4,200 hours.

11. The cost of a manufacturing company for the product is:

Materials	12.00
Labour	9.00
Variable expenses	6.00
Fixed expenses	<u>18.00</u>
Total	<u>45.00</u>

The unit of product is sold for ₹51.00.

The company's normal capacity is 1,00,000 units. The figures given above are for 80,000 units. The company has received an offer for 20,000 units @ ₹36 per unit from a foreign customer.

Advise the manufacturer on whether the order should be accepted. Also give your advice if the order is from a local merchant.

12. The Selling Price per unit of a product is \$ 14. For the forthcoming period, the demand will be only 5,000 units. The fixed expenses at 50% activity (5,000 units) will be \$ 30,000. The company is thinking of shutting down operations, in which case an additional amount of \$ 2,000 will have to be incurred for shutting down and only \$ 20,000 of the above fixed costs can be avoided. What should be the variable cost per unit to recommend a shut-down?
13. G Ltd. produces and sells 95,000 units of X in a year at its 80% production capacity. The selling price of product is \$ 8 per unit. The variable cost is 75% of sale price per unit. The fixed cost is \$ 350,000. The company is continuously incurring losses and management plans to shut-down the plant. The fixed cost is expected to be reduced to \$ 130,000. Additional costs of plant shut-down are expected at \$ 15,000.
- Should the plant be shut-down? What is the capacity level of production of shut-down point?

14. A Paint Manufacturing Company manufactures 200,000 medium-sized tins of 'Spray Lac Paints' annually, when working at normal capacity. Its costs of manufacture per units is \$ 16.40 made up as under:
 Direct Materials = \$ 7.80; Direct Labour = \$ 2.10; Variable Overheads = \$ 2.50 and Fixed Overheads = \$ 4.00.
 Each unit (Tin) of the product is sold for \$ 21 with variable selling expenses of \$ 0.60 per tin. During the next quarter, only 10,000 units can be produced and sold. Management plans to shut down the plant estimating that the fixed manufacturing cost can be reduced to \$ 74,000 for the quarter.
 When the plant is operating, the fixed overheads are incurred at a uniform rate throughout the year. Additional costs of plant shut-down for the quarter are estimated at \$ 14,000. You are required to:
- Express your opinion, along with calculations, as to whether the plant should be shut down during the quarter.
 Calculate the shut-down point for the quarter in units of product (i.e., number of tins).

Break-even & PV Chart:

15. From the following information, prepare the break-even chart.

Fixed Cost = Rs. 2,000
Variable Cost = Rs. 0.50 per unit
Sales = Re. 1 per unit
Units produced and sold 2,000; 4,000; 6,000; 8,000 and 10,000.

16. You are given the following data for the costing year of a factory.

Budget output	1,00,000 units
Fixed Expenses	Rs. 5,00,000
Variable Expenses	Rs. 10 per unit
Selling Price	Rs. 20 per unit

Draw a break-even chart showing the break-even point. If the selling price is reduced to Rs. 18 per unit, what will be the new break-even point?

17. The following figures relate to a particular year's working at 100% capacity level in a manufacturing concern:

	Rs.
Fixed Overheads	1,20,000
Variable overheads	2,00,000
Direct wages	1,50,000
Direct Materials	4,10,000
Sales	10,00,000

Represent the above figures on the Break-even Chart and determine from the chart the break-even point. Verify your results by calculations.

18. Draw the profit volume graph and find out P/V Ratio with the following information.

Output	3,000 units
Volume of Sales	Rs. 7,500
Variable Cost	Rs. 4,500
Fixed Cost	Rs. 1,500

MODULE - 4: RELEVANT COSTING

Meaning of Relevant Cost:

Relevant Costs are costs which are relevant or useful for decision making. They are those expected future costs that are essential but differ for alternative courses of action. They are future costs which would arise as a direct consequence of the decision under review.

Types of Relevant Costing:

1. **Marginal Cost:** Marginal Cost is the total variable cost. It is relevant for decision-making as this will be incurred in future for additional units of production.
2. **Differential Cost:** It is the change in costs due to change in the level of activity or pattern or method of production. When the change results in increase in cost, it is called Incremental Cost, whereas if costs are reduced due to decrease of output, the difference is called Decremental costs.
3. **Opportunity Cost:** It refers to value of sacrifice made or benefit of opportunity foregone by selecting one particular alternative in preference to other alternatives. It is a relevant cost.
4. **Replacement Cost:** It is the cost at which there could be purchase of an asset or material identical to that which is being replaced or re-valued. It is the cost of replacement at current market price and is relevant for decision-making.
5. **Imputed Costs:** These are Notional Costs appearing in the cost accounts only, for example notional rent, interest on capital for which no interest has actually been paid, etc. These costs are relevant for decision making.
6. **Out-of-pocket Cost:** These are costs that entail current or near future cash outlays for the decision at hand. Such costs are relevant for decision. These costs can be avoided or saved if a particular proposal under consideration is not accepted.

Meaning of Irrelevant Costs:

These are costs, which are not relevant or useful for decision-making.

Types of Irrelevant Costs:

1. **Sunk Cost:** It is a cost which has already been incurred or sunk in the past. It is not relevant for decision-making. It is irrelevant because it does not affect the future cash flows of a business.
2. **Committed Cost:** It is a cost in respect of which decision has already been taken, and such decision cannot be altered, e.g. entering into irrevocable agreements for rent, technical collaboration, etc. Committed Costs are not relevant for decision-making.
3. **Absorbed Fixed Cost:** Fixed costs do not change due to increase or decrease in activity. Although such fixed costs are absorbed in cost of production at a normal

rate, they are irrelevant for managerial decision-making. However, if fixed costs are specific, they become relevant.

4. **Non-Cash Expenses:** Non-cash expenses such as depreciation are not relevant because they do not affect the cash flows of a business. However, tax savings on depreciation will be relevant.

PROBLEMS ON RELEVANT COSTING

1. I travel from HSR to Hebbal, twenty five kilometers from my house for my regular work spending Rs. 250 per day. On the way I pick-up my employee Mr. M from Marathahally which is 5 kilometre from HRS. What is the cost of pick-up of my employee Mr. M?
2. ABC Ltd. has been approached by a customer who would like a special job to be done for him, and who is willing to pay Rs. 22,000 for it. The job would require the following materials.

Material	Total unit required	Units already in stock	Book Value of units in stock	Realizable Value	Replacement Cost
A	1,000	0	-	-	Rs. 6 p.u.
B	1,000	600	Rs. 2 p.u.	Rs. 2.50 p.u.	Rs. 5 p.u.
C	1,000	700	Rs. 3 p.u.	Rs. 2.50 p.u.	Rs. 4 p.u.
D	200	200	Rs. 4 p.u.	Rs. 6.00 p.u.	Rs. 9 p.u.

Material B is used regularly by ABC Ltd. and if units B are used for this job, they would need to be replaced to meet other production demand.

Material C and D are in stock as the result of previous over-buying and they have a restricted use. No other use could be found from material C, but the units of Material D could be used in another job as substitute for 300 units of material E, which currently costs Rs. 5 per unit (of which the Company has no units in stock at the moment).

Compute the relevant costs of materials for deciding whether or not to accept the offer.

3. ABC Ltd. is tendering for a six-month contract which would require the use of a specialized machine. The machine was purchased 4 years ago for Rs. 90,000 and now has a Net Book Value of Rs. 35,000. The Company was about to sell

the Machine for Rs. 40,000 but if they used it on this contract, they can sell it after 6 months for Rs. 25,000. The Variable Cost of operating the machine for 6 months would be Rs. 60,000. Ignoring interest costs, identify the Relevant Cost of using the Machine on the contract.

4. Noida Camera Company has received a special order for Photographic Equipment it does not normally produce. The Company has excess capacity, and the order could be manufactured without reducing production of the Firm's regular products. Discuss the relevance of each of the following items in computing the cost of the Special Order:-

- Equipment to be used in producing the order has a Book Value of Rs. 2,000. The Equipment has no other use for Noida Camera Company. If the order is not accepted, the Equipment will be sold for Rs. 1,500. If the Equipment is used in producing the order, it can be sold three months for Rs. 800.
- If the Special Order is accepted, the operation will require some of the storage space in the Company's Plant. If the space is used for this purpose, the Company will rent storage space temporarily in a nearby warehouse at a cost of Rs. 18,000. The Building Depreciation allocated to the Storage Space to be used in producing the special order is Rs. 12,000.
- If the Special Order is accepted, it will require a sub-assembly. Noida Camera can purchase the Sub-assembly for Rs. 24 per unit from an Outside or make it for Rs. 30 per unit. Rs. 30 per unit was determined as follows:

Particulars	Rs.
Direct Material	10.00
Direct Labour	6.00
Variable Overheads	6.00
Apportioned Fixed Overhead	8.00
Total Unit Cost of Sub-assembly	30.00

5. A Company can make any one of the 3 products X, Y or Z in a year. It can exercise its option only at the beginning of each year. Relevant information about the products for the next year is given below:

	X	Y	Z	
Selling Price	Rs. 10 p.u.	Rs. 12 p.u.	Rs. 12 p.u.	
Variable Cost	Rs. 6 p.u.	Rs. 9 p.u.	Rs. 7 p.u.	
Market Demand (units)	3,000	2,000	1,000	
Production Capacity (units)	2,000	3,000	900	
Fixed Cost				Rs. 30,000

You are required to compute the Opportunity Costs for each of the products.

6. A machine manufactures 10,000 units for a part at a total cost of Rs. 21 of which Rs. 18 is variable. This part is readily available in the market at Rs. 19 per unit. If the part is purchased from the market then the Machine can either be utilized to manufacture a component of same quantity contributing Rs. 2 per component or it can be hired out at Rs. 21,000. Recommend which of the alternatives is profitable.
7. S Ltd. is engaged in manufacturing activities. It has received a request from one of its important customers to supply a product which will require conversion of Material M, which is a non-moving item. The following details are available:

Books Value of Material M	Rs. 60
Realizable Value of Material M	Rs. 80
Replacement Cost of Material M	Rs. 100

It is estimated that conversion of one unit of M into one unit of the Finished Product will require 1 labour hour. At present, labour is paid at the rate of Rs. 20 per hour. Other Costs are as follows:

Out-of-pocket expenses	Rs. 30 per unit
Allocated Overheads	Rs. 10 per unit

The labour will be re-deployed from other activities. It is estimated that the temporary redeployment will not result in loss of contribution. The employees to be re-deployed are permanent employees of the company.

Estimate the Minimum Price to be charged from the customer so that the Company is not worse off by executing the order.

8. A machinery was purchased for Rs. 20,00,000 the operating cost is Rs. 100 per hour. When the machine was about to be installed it was found out that another machine which is more efficient was available in the market for Rs. 35,00,000. The operating cost of the machine being Rs.70 per hour. If this new machine is purchased for Rs. 35,00,000 the old machine could be disposed off for Rs. 12,00,000. Consider the revenue per hour from both the machines as Rs. 120.
- Identify the relevant cost
 - How many hours does the new machine have to run for the benefit of the company?
 - What is the break-even hour for both these machines?
9. 20,000 shirts have been manufactured for export at a cost of Rs.150 per shirt. These shirts were rejected because of defects and can be sold for Rs.120 per shirt

in the local market. If the defects are rectified by spending Rs.30 per shirt, the same can be sold for Rs.160.

- i) What are the relevant costs
- ii) Should the rectification be done
- iii) What is the gain or loss if not rectified
- iv) What is the gain or loss if rectified

10. A printer has received a special order for the printing of a brochure. A brochure requires special type of paper that is not regularly used. A limited quantity of paper (100 reams) is remaining from the previous job completed in the previous year. The cost of the paper last year was 15 dollars per ream.

Brochure requires 250 reams, current market price is 26 dollars per ream. The resale value of the paper in inventory is 10 dollars per ream. Identify the relevant cost of the paper used in the printing of the brochure.

- a) What are the relevant costs in the problem given?
- b) Calculate the total relevant cost?

11. 10,000 shirts have been manufactured for export at a cost of Rs. 300 per shirt. These shirts were rejected because of defects and can be sold for Rs. 240 per shirt in the local market. If the defects are rectified by spending Rs. 60 per shirt, the same can be sold for Rs. 320.

- i) What are the relevant costs?
- ii) Should the rectification be done?
- iii) What is the gain or loss if not rectified?
- iv) What is the gain or loss if rectified?

12. A machinery was purchased for Rs. 26,00,000. The operating cost is Rs. 130 per hour. When the machine was about to be installed it was found out that another machine which is more efficient was available in the market for Rs. 45,50,000. The operating cost of the machine being Rs. 91 per hour.

If this new machine is purchased for Rs. 45,50,000, the old machine could be disposed off for Rs. 15,60,000. Consider revenue per hour from the machine as Rs. 195.

- i) Identify the relevant cost.
- ii) How many hours does the new machine have to run for the benefit of the company?
- iii) Calculate the break-even hours of both the machines.

13. The cost of making component Q, which forms part of Product Y, is given below:

Particulars	Cost per unit in Rs.
-------------	----------------------

Raw Materials	4
Direct Labour	8
Production Overheads (60% Variable)	16
Total	28

Component Q could be bought from an outside supplier for Rs. 20 per unit. State whether the company should make or buy component Q.

14. The selling price of Product X is Rs. 22 and production cost for one unit is:

Particulars	Cost per unit in Rs.
Raw Materials	8
Direct Labour	4
Production Overheads	8
Total	20

Variable production overhead which is part of production overhead is calculated at 50% of labour cost. There is a possibility of supplying a special order for 2000 units of product X at Rs. 16 each. If this order is accepted, the normal budgeted sales would not get affected and the company has the necessary capacity to produce the additional units. State whether the company should accept the special order.

15. The production manager of your organization has approached you for some costing advice on project X, a special order from overseas for Rs. 30,000. The costs associated with the project are as follows:

Particulars	Cost in Rs.
Raw Material - A	4,000
Raw Material - B	8,000
Direct Labour	6,000
Supervision	2,000
Overheads	12,000
Total	32,000

You ascertain the following:

- i) Material A is in stock and the above was the cost. This Material is of no use to the organization other than the above project. If this material is not used, it could be realized at Rs. 1,750.
 - ii) The organization has no stock of material B. The required material for the project has to be ordered at the cost shown above.
 - iii) Direct labour cost of Rs. 6,000 related to workers that will be transferred to this project from another project. Extra labour will need to be recruited to the other project at a cost of Rs. 7,000.
 - iv) Supervision costs have been charged to the project on the basis of 1/3 of labour cost and will be carried out by the existing staff within their normal duty.
 - v) Overheads have been charged to the project at the rate of 200% on direct labour.
 - vi) A new machinery has to be purchased specially for the project at a cost of Rs. 8,000 and then disposed off at Rs. 5,000 at the end of the project.
- Required:
- a) Find the out the relevant costs for the special order whether to accept or to reject the special order.
 - b) Should the special order be accepted or rejected?

MODULE 5: BUDGET AND BUDGETARY CONTROL

Introduction: Every business enterprise needs the use of control techniques for surviving in the highly competitive and changing economic world. There are various control devices in use. Budgets are the most important tool of profit planning and control. They also act as an instrument of co-ordination.

Budget: “A budget is a pre-determined statement of management policy during a given period which provides a standard for comparison with the results actually achieved”. – Crown and Howard.

Budgetary Control: “It is a system of controlling costs which includes the preparation of budgets, coordinating the department and establishing responsibilities, comparing actual performance with the budgeted and acting upon results to achieve maximum profitability.” - Brown and Howard.

Budgetary Control involves the following functions:

- i) The objects are set by preparing budgets.
- ii) The business is divided into various responsibility centers for preparing various budgets.
- iii) The actual figures are recorded.
- iv) The budgeted and actual figures are compared for studying the performance of different cost centers.
- v) If actual performance is less than the budgeted norms, a remedial action is taken immediately.

Thus, the three cardinal features of budgetary control are Planning, Coordinating and Control.

A budget is a blue print of a plan expressed in quantitative terms. Budgeting is a technique for formulating budgets. Budgetary control, on the other hand, refers to the principles, procedures and practices of achieving given objectives through budgets.

Requirements of a Sound Budgetary Control System:

1. **Clarifying Objectives:** The budgets are used to realize objectives of the business. The objectives must be clearly spelt out so that budgets are properly prepared. In the absence of clear goals, the budgets will also be unrealistic.
2. **Proper Delegation of Authority and Responsibility:** Budget preparation and control is done at every level of management. Even though budgets are finished at top level but involvement of persons from lower levels of management is essential for their process. This necessitates proper delegation of authority and responsibility.

3. **Proper Communication:** An effective system of communication is required for a successful budgetary control. The flow of information regarding budgets should be quick so that these are implemented. The upward communication will help in knowing the difficulties in implementation of budgets. The performance reports of various levels will help top management in budgetary control.
4. **Budget Education:** The employees should be properly educated about the benefits of budgeting system. They should be educated about their role in the success of this system. Budgetary control may not be taken only as a control device by the employees but it should be used as a tool to improve their efficiency.
5. **Participation of all Employees:** Budgeting is done for every segment of the business. It will also require the active participation and involvement of all employees. In practice the budgets are to be executed at lower levels of management. Those for whom the budgets are framed should be actively associated with their preparation and execution. The employees, on the basis of their past experience, may give more practical and useful suggestions. The success of budgetary control system depends upon the participation of all employees of their organization.
6. **Flexibility:** Flexibility in budgets is required to make them suitable under changed circumstances. Budgets are prepared for the future, which is always uncertain. Even though budgets are prepared by considering the future possibilities but still some occurrences later on may necessitate certain adjustments. Flexibility will make the budgets more appropriate and realistic.
7. **Motivation:** Budgets are to be implemented by human beings. Their successful implementation will depend upon the interest shown by the employees. All persons should be motivated to improve their working so that budgeting is successful. A proper system of motivation should be introduced for making this system a success.

Advantages of Budgetary Control:

1. **Maximisation of Profit:** The Budgetary control aims at maximization of profits. To achieve this aim, a proper planning and co-ordination of different functions is undertaken. There is a proper control over various capital and revenue expenditures. The resources are put to the best possible use.
2. **Co-ordination:** The working of different departments and sectors is properly co-ordinated. The budgets of different departments have a bearing on one another. The co-ordination of various executives and subordinates is necessary for achieving budgeted targets.
3. **Specific Aims:** The plans, policies and goals are decided by the top management. All efforts are put together to reach the common goal of the organization. Every department is given a target to be achieved. The efforts are directed towards

achieving some specific aims. If there is no definite aim then the efforts will be wasted in pursuing different aims.

4. **Tool for measuring performance:** By providing targets to various departments, budgetary control provides a tool for measuring managerial performance. The budgeted targets are compared to actual results and deviations are determined. The performance of each department is reported to the top management. This system enables the introduction of management by exception.
5. **Economy:** The planning of expenditure will be systematic and there will be economy in spending. The finances will be put to optimum use.
6. **Determining weaknesses:** The deviations in budgeted and actual performance will enable the determination of weak spots. Efforts are put on those aspects where performance is less than the stipulated.
7. **Corrective action:** The management will be able to take corrective measures whenever there is a discrepancy in performance. The deviations will be regularly reported so that necessary action is taken at the earliest.
8. **Consciousness:** By fixing targets for the employees, they are made conscious of their responsibility.
9. **Reduces Costs:** In the present day competitive world budgetary control has a significant role to play. Every businessman tries to reduce the cost of production for increasing sales.
10. **Introduction of Incentive Schemes:** Budgetary control system also enables the introduction of incentive schemes of remuneration. The comparison of budgeted and actual performance will enable the use of such schemes.

Limitations of Budgetary Control:

1. **The Budget Plan is based on estimates:** Budgets are based on forecasts. Absolute accuracy is not possible in forecasting and budgeting.
2. **Danger of rigidity:** A budget program must be dynamic and continuously deal with the changing business conditions. If budgets are not revised with the changing circumstances, they will become rigid.
3. **Budgeting cannot replace management:** Budgeting is only a tool of management.
4. **Opposition from staff:** Employees may not like to be evaluated and thus oppose introduction of budgetary control system.
5. **Expensive technique:** The installation and operation of a budgetary control system is a costly affair as it requires the employment of specialized staff and involves additional expenditure.

Classification of Budgets:

- A) **On the basis of functions:** They are components of Master Budget.

Sales Budget	Production Budget
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Production Cost Budget	Raw Material Budget
Purchases Budget	Labour Budget
Production Overhead Budget	Selling and Distribution Budget
Administration Cost Budget	Cash Budget
Capital Expenditure Budget	

B) On the basis of level of capacity or capacity utilization:

Fixed Budget	Flexible Budget
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Fixed Budget: It is a static budget that is based on the projected level of output prior to the start of the period. In other words, a fixed budget is one which is prepared keeping in mind one level of output. It is prepared on the assumption that output and sales can be estimated with a fair degree of accuracy.

Flexible Budget: A flexible budget is one which adjusts the static (fixed) budget for the actual level of output. It is a budget which is designed to change in relation to the level of activity attained.

Distinction between Fixed and Flexible Budgets:

Fixed Budget	Flexible Budget
It assumes static business conditions.	It assumes changing business conditions.
It is prepared for only one level of activity.	It is prepared for different level of activity.
Fixed budget figures are not changed when actual level of activity changes.	In flexible budget, the figures are adjusted according to the actual level of activity attained.
Meaningful comparison is not possible when actual level and budgeted level are not the same.	Meaningful comparison is possible even when actual level and budgeted level are not the same.
They are not very useful under changing business environment.	They are very useful under changing business environment.

Problems on Budgetary Control:

1. The expenses for the production of 5,000 units in a factory are given as follows:

Particulars	Per Unit (Rs.)
Materials	50
Labour	20
Variable Overheads	15
Fixed Overheads (Rs. 50,000)	10
Administrative expenses (5% variable)	10
Selling Expenses (20% Fixed)	6
Distribution expenses (10% Fixed)	5
Total cost of sales per unit	116

You are required to prepare a budget for the production of 7,000 units.

2. The following information at 50% capacity is given. Prepare a flexible budget and forecast the profit or loss at 60%, 70% and 90% capacity.

Particulars	Expenses @ 50% capacity
Fixed Expenses:	
Salaries	50,000
Rent and Taxes	40,000
Depreciation	60,000
Administrative Expenses	70,000
Variable Expenses	
Materials	2,00,000
Labour	2,50,000
Others	40,000
Semi-variable Expenses:	
Repairs	1,00,000
Indirect Labour	1,50,000
Others	90,000

It is estimated that fixed expenses will remain constant at all capacities. Semi-variable expenses will not change between 45% and 61% capacity, will rise by 10% between 61% and 75% capacity, a further increase of 5% when capacity crosses 75%.

Estimated sales at various levels of capacity are:

Capacity	Sales (Rs.)
60%	11,00,000
70%	13,00,000
90%	15,00,000

3. The following information relates to a flexible budget at 60% capacity. Find out the overhead costs at 50% and 70% capacity and also determine the overhead rates:

Particulars	Expenses @ 60% capacity
Variable overheads:	
Indirect Labour	10,500
Indirect Materials	8,400
Semi-variable Overheads:	
Repairs and Maintenance (70% fixed, 30% variable)	7,000
Electricity (50% fixed, 50% variable)	25,200
Fixed Overheads:	
Office expenses including salaries	70,000
Insurance	4,000
Depreciation	20,000
Estimated direct labour hours	1,20,000

4. With the following data for a 60% activity, prepare a budget for production at 80% and 100% capacity:

Production at 60% activity	600 units
Materials	Rs. 100 per unit
Labour	Rs. 40 per unit
Direct Expenses	Rs. 10 per unit
Factory overheads	Rs. 40,000 (40% fixed)
Administrative Expenses	Rs. 30,000 (60% fixed)

5. A company is expecting to have Rs. 32,000 cash in hand on 1.4.2008 and it requests you to prepare cash budget for the three months, April to June 2008. The following information is supplied to you:

Month	Sales (Rs.)	Purchases (Rs.)	Wages (Rs.)	Expenses (Rs.)
Feb	70,000	44,000	6,000	5,000
Mar	80,000	56,000	9,000	6,000
Apr	96,000	60,000	9,000	7,000
May	1,00,000	68,000	11,000	9,000
Jun	1,20,000	62,000	14,000	9,000

Other information:

- Period of credit allowed by suppliers is two months.
- 25% of sales are for cash and the period of credit allowed to customers for credit sales is one month.
- Delay in payment of wages and expenses one month.
- Income tax Rs. 28,000 is to be paid in June 2008.

6. XY Co. wishes to arrange overdraft facilities with bankers during the period April to June of a particular year, when it will be manufacturing mostly for stock. Prepare a Cash budget for the above period from the following data, indicating the extent of the bank facilities, the company will require at the end of each month:

a)

Month	Sales (Rs.)	Purchases (Rs.)	Wages (Rs.)
Feb	1,80,000	1,24,800	12,000
Mar	1,92,000	1,44,000	14,000
Apr	1,08,000	2,43,000	11,000
May	1,74,000	2,46,000	10,000
Jun	1,26,000	2,68,000	15,000

b) 50% of the credit sales are realized in the month following the sales and the remaining sales in second month following.

c) Creditors are paid in the following month of purchase.

d) Cash at bank on 1st April Rs. 25,000.

7. From the following budget data, forecast the cash position at the end of April, May and June 2016:

Month	Sales (Rs.)	Purchases (Rs.)	Wages (Rs.)	Misc. Exps (Rs.)
Feb	1,20,000	84,000	10,000	7,000
Mar	1,30,000	1,00,000	12,000	8,000
Apr	80,000	1,04,000	8,000	6,000
May	1,16,000	1,06,000	10,000	12,000
Jun	88,000	80,000	8,000	6,000

Additional Information:

Sales : 20% realized in the month of sales, discount allowed 2%. Balance realized equally in two subsequent months.

Purchases: These are paid in the month following the month of supply.

Wages: 25% paid in arrears in the following month.

Miscellaneous Expenses: Paid a month in arrears.

Rent: Rs. 1,000 per month paid quarterly in advance due in April.

Income Tax: First installment of advance tax Rs. 25,000 due on or before 15th June.

Income from Investments: Rs. 5,000 received quarterly in April, July, etc.

Cash in hand: Rs. 5,000 on 1st April, 2016.

8. A company is drawing its production plan for the year 2002-03 in respect of two of its products 'Gamma' and 'Delta'. The company's policy is not to carry any closing WIP at the end of any month. However, its policy is to hold a

closing stock of finished goods at 50% of the anticipated quantity of sales of the succeeding month. For the year 2002-03 the company's budgeted production is 20,000 units of 'Gamma' and 25,000 'Delta'. The following is the estimated cost data:

	Gamma	Delta
Direct Material per unit	Rs. 50	Rs. 80
Direct Labour per unit	Rs. 20	Rs. 30
Other manufacturing expenses apportionable to each type of product based on production.	Rs. 2,00,000	Rs. 3,75,000

The estimated units to be sold in the first 7 months of the year 2002-03 are as under:

	April	May	June	July	Aug.	Sept.	Oct.
Gamma	900	1100	1400	1800	2200	2200	1800
Delta	2900	2900	2500	2100	1700	1700	1900

You are required to:

- Prepare a production budget showing month-wise number of units to be manufactured.
 - Present a summarized production cost budget for the half-year ending 30-09-2002
9. A Company produces and sells three items: (a) Snow Cream, (b) Talcum Powder and (c) Cold Cream. The Company has divided its market into two Zones: Zone A and Zone B. The actual figures for the previous year sales were as under:

	Zone A		Zone B	
	Units	Unit Price (Rs.)	Units	Unit Price
Snow Cream	4,00,000	12.00	2,50,000	12.00
Talcum Powder	2,50,000	15.00	3,50,000	15.00
Cold Cream	3,00,000	16.00	3,00,000	16.00

For the Current Year, it is estimated that sale of snow cream will go up by 10% in Zone B and of Cold Cream by 25,000 units in Zone A. The company plans to introduce a publicity film for Talcum Powder in the T.V. network. The budgeted figures for Talcum Powder are to be increased by 20% in both the Zones.

The prices of the two creams are to be maintained but for Talcum Powder, a bonus cut of Re. 1 will be announced.

You are required to prepare quantitative-cum-financial budget for sales in the current year.

10. From the following particulars, prepare a Production Budget of a Company for the year ended June30, 2010.

Product	Sales in units (as per Sales Budget)	Estimated Stock in units	
		1 July, 2009	30 June 2010
A	1,50,000	14,000	15,000
B	1,00,000	5,000	14,500
C	70,000	8,000	8,000

11. Draw a **material procurement budget (quantitative)** from the following information:

Estimated sales of a product are 40,000 units. Each unit of the product requires 3 units of material A and 4 units of material B.

Estimated opening balances at the commencement of the next year:

Particulars	Units
Finished Products	5,000
Material A	12,000
Material B	20,000

Materials on order:

Particulars	Units
Material A	7,000
Material B	11,000

The desirable closing balances at the end of the next year are:

Particulars	Units
Finished Products	7,000
Material A	15,000
Material B	25,000

Materials on order:

Particulars	Units
Material A	8,000
Material B	10,000

12. The Sales Director of a manufacturing Company reports that next year he expects to sell 50,000 units of a particular product:

The Production Manager consults the store-keeper and casts his figures as follows:

Two kinds of raw materials A and B are required for manufacturing the products.

Each unit of the product requires 2 units of A and 3 units of B.

The estimated opening balances at the commencement of the next year are:

Particulars	Units
Finished Products	10,000
Material A	12,000

Material B	15,000
------------	--------

The desirable closing balances at the end of the next year are:

Particulars	Units
Finished Products	14,000
Material A	13,000
Material B	16,000

Draw up a quantitative chart showing materials purchase budget for the next year.

13. The budget manager of Cosmetics Ltd., is preparing a budget for the accounting year starting from 1st July.

As part of the budget operations, some items of factory overhead costs have been estimated by him under specified conditions of volume as follows:

Volume of Production (units)	1,20,000	1,50,000
Expenses:	Rs.	Rs.
Indirect Materials	2,64,000	3,30,000
Indirect Labour	1,50,000	1,87,500
Maintenance	84,000	1,02,000
Supervision	1,98,000	2,34,000
Engineering Service	94,000	94,000

Calculate the cost of factory overhead items given above at 1,40,000 units of production.

Master Budget

14. A company requires you to prepare Master Budget from the following:

Sales per annum:

Product A- Rs. 40,00,000

Product B- Rs. 60,00,000

Other information:

Direct Material cost = 40% of sales

Direct Wages: 15 workers @ Rs. 12,000 per month per worker.

Factory Overheads: Works manager salary at Rs. 20,000 per month, Foreman's salary at Rs.15,000 per month.

Stores & Spares: 2½ % on sales, depreciation on machinery is Rs. 1,26,000 per annum.

Light and power: Rs.50,000 p.a.; Repairs and Maintenance: Rs. 80,000 per annum; Miscellaneous expenses: 10% of Direct Wages; Administration, selling and distribution expenses: Rs. 1,40,000 p.a.

15. A company manufactures three products A, B & C and sales at Rs. 450, Rs. 550 & Rs. 650 per unit respectively. The ratio of sales in quantity of A, B & C is 1:2:4. Other information for the three products-

Product	Quantity of parts required (nos.)				Labour hours		Variable overheads p. u produced (Rs.)
	D	E	F	G	Skilled	unskilled	
A	1	10	2	8	6	8	9
B	1	2	14	10	4	6	11
C	1	6	10	2	3	6	7

The present purchase price per part is Rs. 45, Rs. 15, Rs. 15 & Rs. 5 for D, E, F & G respectively. The wages rate per hour is skilled worker Rs. 6 and unskilled Rs. 5

The opening stock is 500, 1,000, 3,000, 1,500, 1,000, 20,000 & 10,000 for A, B, C, D, E, F & G. The closing stock of the product and the parts is 90% of the opening stock.

The workers work for 8 hours a day for 25 days in a month. The share of fixed overhead **per month** for A, B & C comes to Rs. 15,75,000, Rs. 5,80,000 & Rs. 8,45,000 respectively. The yearly profit projected is Rs. 1,20,00,000.

Prepare the following budgets for the next **fiscal month**.

- Sales budget in quantity and value
- Production budget
- Parts usage budget
- Purchase budget in quantity and value
- Man power budget showing labour hours and wages payable
- Master budget

MODULE 6: STANDARD COST AND VARIANCE ANALYSIS

Standard cost is a predetermined estimate of cost to manufacture a single unit or a number of units of a product during a future period.

Standard Costing:

According to Bigg, “Standard Costing discloses the cost of deviations from standards and clarifies these as to their causes, so that management is immediately informed of the sphere of operations in which remedial action is necessary”.

According to I.C.M.A. Terminology, Standard Costing is “the preparation and use of standard costs, their comparison with actual costs and the analysis of variances to their causes and points of incidence”.

The technique of standard cost study comprises of:

- a) Pre-determination of standard costs.
- b) Use of standard costs
- c) Comparison of actual cost with the standard cost
- d) Find out and analyse reasons for variances
- e) Reporting to management for proper action to maximize efficiency.

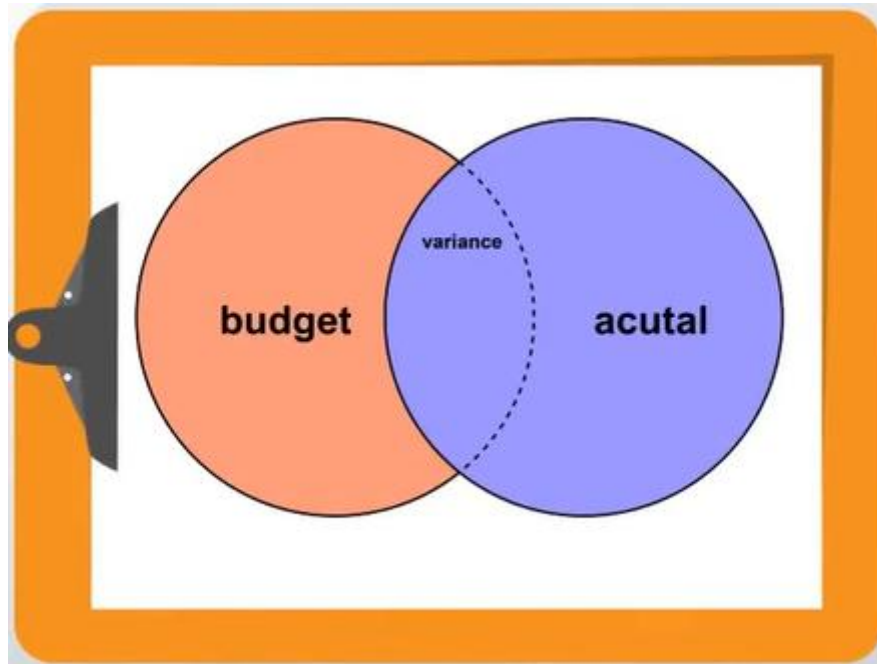
Advantages of Standard Costing:

- a) It helps the management in formulating price and production policy.
- b) It is a yardstick of performance.
- c) It reduces avoidable wastage and losses.
- d) Executives become more responsible, as it shows clearly who is responsible for the cost centres.
- e) It aids in budgetary control and in decision-making.
- f) It creates cost consciousness among the personnel, because the variance analysis fixes responsibility for favourable or unfavourable performances.

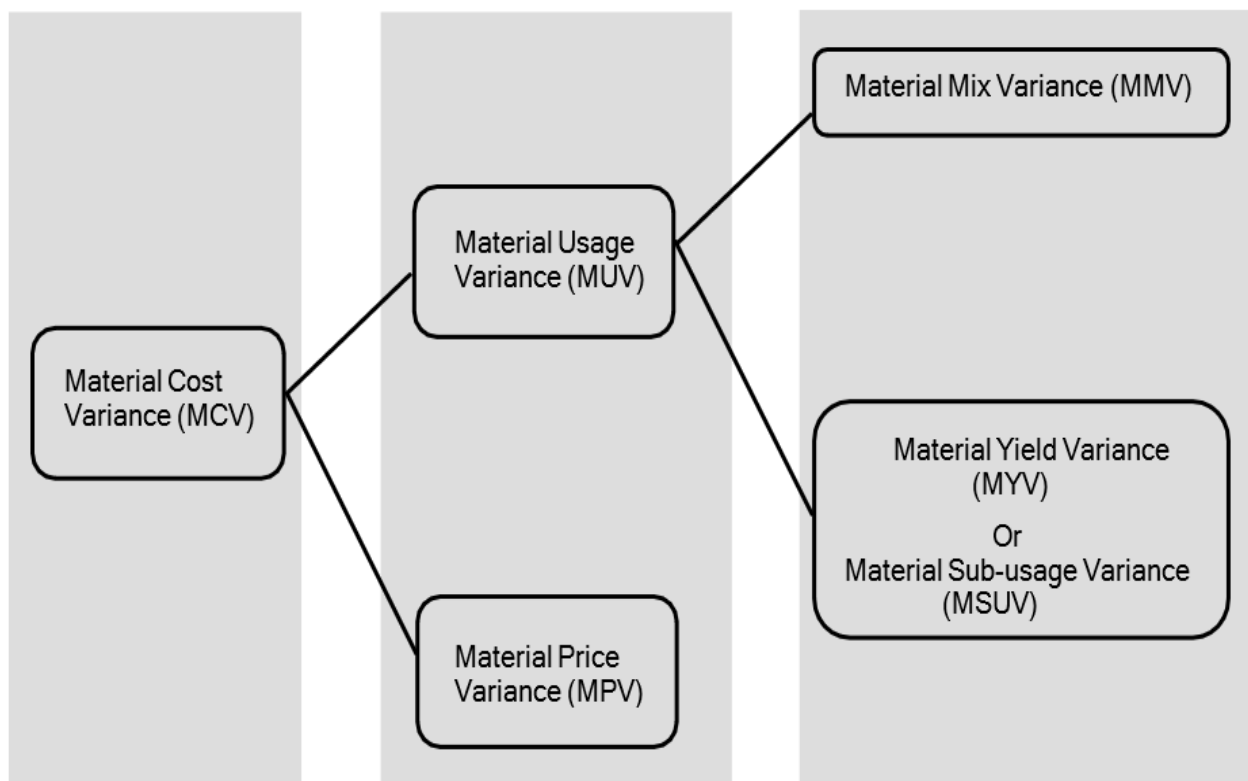
Disadvantages of Standard Costing:

- a) It is costly, as the setting of standards needs high technical skill.
- b) Keeping of up-to-date standard is a problem. Periodic revision of standard is a costly affair.
- c) Since it is difficult to set correct standards, it is difficult to ascertain correct variance.
- d) For small concerns, standard costing is expensive.
- e) Standard costing may not be effective in industries which deal in non-standardised products or jobs according to customer's requirements.

Computation of Variances:



Material Variances



1. **Material Cost Variance:** It is the difference between the standard cost of direct materials specified for the output achieved and the actual cost of direct material used.

Material Cost Variance = Standard Cost of Material for Actual Output – Actual Cost of Materials Used

OR

$$(SQ \times SP) - (AQ \times AP)$$

2. **Material Price Variance:** Material Price Variance is that portion of the direct material cost variance which is the difference between the standard price specified and the actual price paid for the direct materials used.

Material Price Variance = Actual Quantity (Standard unit price – Actual unit price)

OR

$$AQ (SP - AP)$$

3. **Material Usage (Quantity) Variance:** It is that portion of the direct material cost variance which is the difference between the standard quantity specified for the production achieved, whether completed or not, and the actual quantity used, both valued at standard price.

Material Usage Variance = Standard Price (Actual Quantity – Standard Quantity)

OR

$$SP (AQ - SQ)$$

4. **Material Mix Variance:** It is that portion of direct material usage variance which is the difference between the actual quantities of ingredients used in a mixture at standard price and the total quantity of ingredients used at the weighted average price per unit of ingredients as shown by the standard cost sheet.

Material Mix Variance = Standard Price (Revised Standard Quantity – Actual Quantity)

OR

$$SP (RSQ - AQ)$$

5. **Material Yield Variance:** It is that portion of the Direct Material Usage Variance which is due to the difference between the standard yield specified and the actual yield obtained.

Material Yield Variance = Standard cost per unit (Actual yield – Standard yield)

OR

SC p.u. (AY-SY)

Note: AY will never change. SY will be calculated for the actual mix of quantity as under:

New SY = (Old SY/TSQ) TAQ

STEPS IN CALCULATING MATERIAL COST VARIANCE

FIRST STEP:

M1 = Actual Cost of Actual Material used (AQ x AR)

M2 = Standard Cost of Actual Material used (AQ x SR)

M3 = Standard Cost of Actual Material used, if it had been in Standard proportion.
(AQ x Std. Mix x SR)

M4 = Standard Cost of Actual Output (SQ x SR)

Where AQ = Actual Quantity; AR = Actual Rate; SQ = Standard Quantity and SR = Standard Rate.

SECOND STEP:

Material Price or Rate Variance =	M1 – M2
Material Mix Variance =	M2 – M3
Material Yield Variance =	M3 – M4
Material Usage (Quantity) Variance =	M2 – M4
Material Cost Variance =	M1 – M4

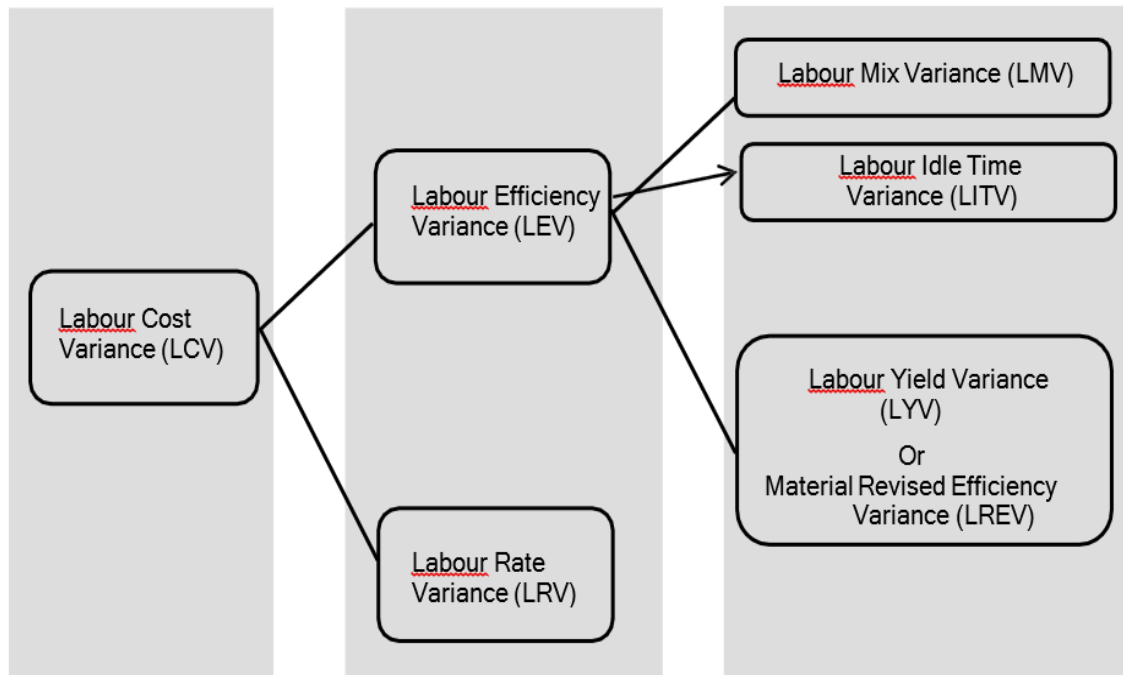
THIRD STEP:

RECONCILIATION:

Material Cost Variance = Material Price Variance + Material Usage (Quantity) Variance

Material Usage Variance = Material Mix Variance + Material Yield Variance

Labour Variances



1. **Labour Cost Variance:** It is the difference between the Actual wages paid and the standard wages specified for the production.

$$\text{LCV} = (\text{Standard Hours} \times \text{Standard Rate}) - (\text{Actual Hours} \times \text{Actual Rate})$$

OR

$$(\text{SH} \times \text{SR}) - (\text{AH} \times \text{AR})$$

2. **Labour Rate Variance:** This variance is the direct result of the wages paid at a rate different from the standard rate. That is, it is the difference between the standard rate of pay specified and the actual rate paid.

$$\text{Labour Rate Variance} = \text{Actual Hours} (\text{Standard Rate} - \text{Actual Rate})$$

OR

$$\text{AH} \times (\text{SR} - \text{AR})$$

3. **Labour Time or Efficiency Variance:** It is the difference between the actual hours paid and the standard hours allowed for the actual output achieved.

$$\text{Labour Efficiency Variance} = \text{Standard Wage Rate} (\text{Standard Hours of Production} - \text{Actual Hours Worked})$$

OR

$$\text{SR} \times (\text{SH} - \text{AHW})$$

4. **Idle Time Variance:** This type of variance arises because of the time during which the labour remains idle due to abnormal reasons, i.e., power failures, machine breakdown, shortage of materials, etc. Idle Time Variance is always adverse. Abnormal Idle Time Variance is the product of Abnormal idle time and the standard hourly rate.

Labour Idle Time Variance (LITV) = (Actual hours paid for x Standard rate) – (Actual hours worked x Standard rate)

OR

Idle Hours x Standard rate.

5. **Labour Mix Variance or Labour Gang Variance** :It is the difference between the actual composition of a gang and the standard composition specified for a gang.

Labour mix variance = (Actual hours at standard rate of actual gang – Actual hours at standard rate of standard gang)

OR

Standard rate (Revised standard labour hours - Actual labour hours)

OR

LMV = (RSH - AHW) x SR

6. **Labour Yield Variance:** This variance arises due to difference between actual output of worker and standard output of worker specified.

Labour yield variance = Standard labour cost unit (Actual output – Standard output)

OR

(Standard loss of actual total input – Actual loss) x Average standard rate per unit.

OR

LYV = SC p.u. (AY – SY)

STEPS IN CALCULATING LABOUR COST VARIANCE

FIRST STEP:

L1 = Actual Payment to workers for Actual hours worked (AH × AR)

L2 = Standard Cost of Actual hours worked (AH × SR)

L3 = Standard Cost of Actual labour hours worked, if it had been in Std. proportion.
(AH × Std. Mix × SR)

L4 = Standard Cost of Actual labour hours utilized. (AH Utilised × Std. Mix × SR)

L5 = Standard Cost of Actual Output (Std. Cost of Std. output / Std. output × Actual output)

Where AH = Actual Hours; AR = Actual Rate; SH = Standard Hours and SR = Standard Rate.

SECOND STEP: Computation of Variances:

Labour Rate Variance	L1 - L2
Labour Mix Variance or Labour Gang Variance	L2 - L3
Labour Idle Time Variance	L3 - L4
Labour Yield Variance	L4 - L5
Labour Efficiency Variance	L2 - L5

THIRD STEP: RECONCILIATION:

Labour Cost Variance = Labour Rate Variance + Labour Efficiency Variance

Labour Efficiency Variance = Labour Mix Variance + Labour Idle Time Variance + Labour Yield Variance

Overhead Cost Variance

The total overhead cost variance is the difference between the standard cost of overhead allowed for the actual output achieved and the actual overhead cost incurred. In other words, overhead cost variance is the under or over-absorption of overheads.

However, before proceeding to study these variances, one should be aware of the basic terms used in the computation of overhead variance:

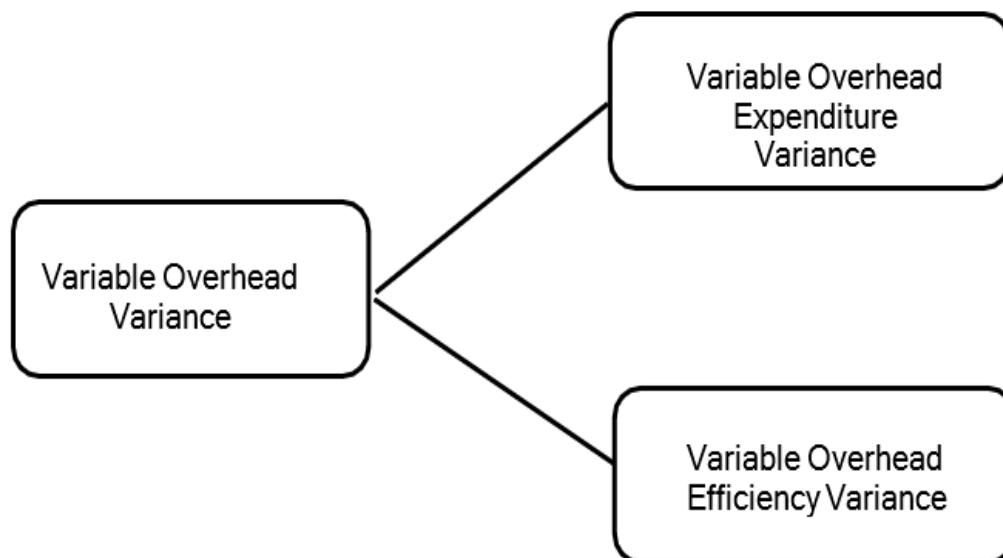
- (i) Standard overhead rate (per unit) = $\frac{\text{Budgeted overhead}}{\text{Budgeted output in units}}$
- (ii) Standard overhead rate (per hour) = $\frac{\text{Budgeted overhead}}{\text{Budgeted hours}}$
- (iii) Standard hours for actual output = $\frac{\text{Budgeted hours} \times \text{Actual output}}{\text{Budgeted output}}$
- (iv) Standard output for actual hours = $\frac{\text{Budgeted output (in units)} \times \text{Actual hours}}{\text{Budgeted hours}}$
- (v) Absorbed (or Recovered) overhead = Standard Rate per hour $\times$ Actual Output
Or standard rate per unit $\times$ standard hours for actual output
- (vi) Budgeted overhead = Budgeted output $\times$ Std. overhead rate per unit
Or Budgeted hours $\times$ Std. overhead rate per hour
- (vii) Standard overhead = Std. output for actual time $\times$ Std. overhead rate per unit
Or Actual hours $\times$ Std. overhead rate per hour
- (viii) Actual overhead = Actual output $\times$ Actual overhead rate per unit
Or Actual overhead = Actual output $\times$ Actual overhead rate per unit

Overhead cost variance = [Actual Output $\times$ Standard Overhead Rate Per Unit] – Actual Overhead Cost

OR

[Standard Hours for Actual Output $\times$ Standard Overhead Rate Per Hour] – Actual Overhead Cost

Variable Overhead Variances



Variable Overhead Variance: Variable Overhead Variance is the difference between Actual overhead incurred during the period and standard variable overhead for production.

Variable Overhead Variance = (Standard Variable Overhead Rate × Actual Output) - Actual Variable Overheads

OR

(Standard Hours for Actual Output × Standard Variable Overhead Rate) - Actual Variable Overheads

OR

(Standard Rate × Actual output) - (Actual Rate × Actual output)

Variable Overhead Expenditure Variance: Variable Overhead variance is the difference between the Actual Variable Overhead and Standard Variable Overhead appropriate to the level of activity attempted.

Variable Overhead Expenditure Variance (VOExV) = (Actual Hours × Standard Variable Overhead Rate per Hour) - Actual Variable Overhead

OR

Actual Hours (Standard Variable Overhead Rate per Hour - Actual Variable Overhead Rate per Hour)

Variable Overhead Efficiency Variance: It is the difference between standard variable overhead rate and standard variable overhead for the production.

Variable Overhead Efficiency Variance (VOEfV) = (Standard Time for Actual Production × Standard Variable Overhead Rate per Hour) - Actual Hours Worked × Standard Variable Overhead Rate per Hour).

OR

Standard Variable Overhead on Actual Production - Standard Variable Overhead for actual time.

OR

Recovered Overheads - Standard Overheads

STEPS IN CALCULATING VARIABLE OVERHEAD VARIANCE

FIRST STEP:

VO1 = Actual Variable overhead incurred during the period (normally given)

VO2 = Standard Variable Cost for Actual hours worked (AH x SR)

VO3 = Standard Variable Overhead for the Actual Production (Std. Variable overhead per unit x Actual Production)

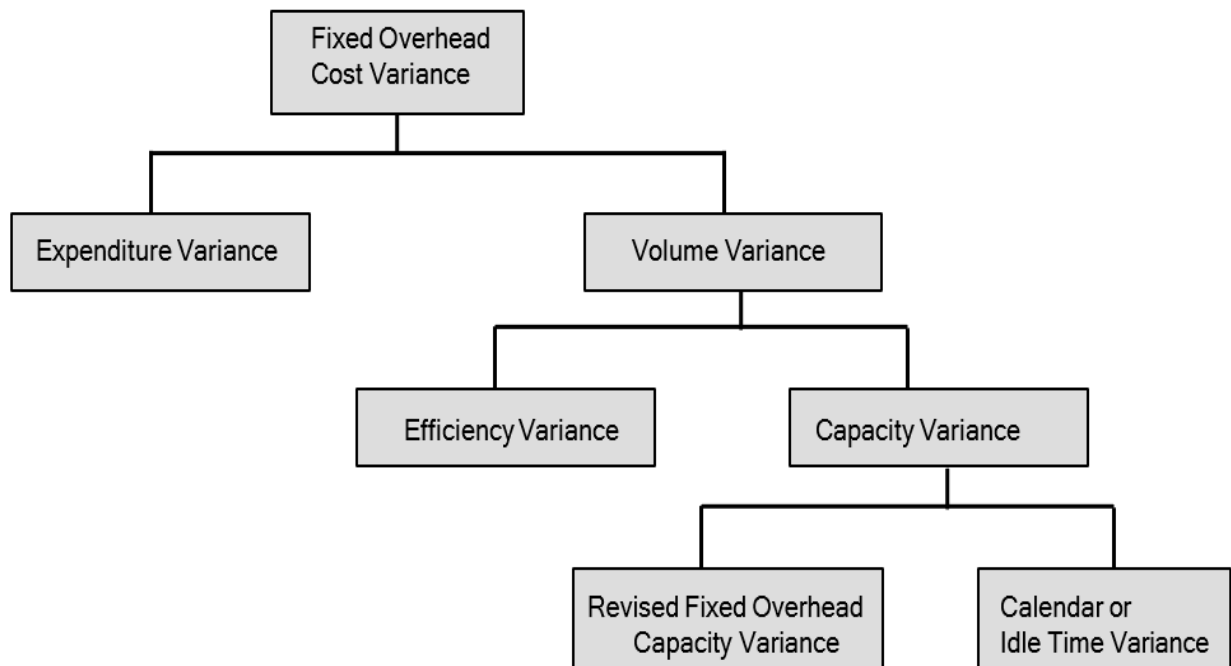
SECOND STEP: Computation of Variances:

Variable Overhead Expenditure Variance	VO1 - VO2
Variable Overhead Efficiency Variance	VO2 - VO3
Variable Overhead Variance	VO1 - VO3

THIRD STEP: RECONCILIATION:

Variable Overhead Variance = Variable Overhead Expenditure Variance + Variable Overhead Efficiency Variance

Fixed Overhead Variances



Fixed Overhead Variance: Fixed Overhead Variance is the difference between Actual Fixed Overhead incurred during the period and standard fixed overhead for production.

$\text{FOCV} = (\text{Actual output} \times \text{Standard fixed overhead rate}) - \text{Actual fixed overheads}$

OR

$(\text{Standard hours produced} \times \text{Standard fixed overhead rate per hour}) - \text{Actual fixed overheads}$

OR

$\text{Recovered fixed overhead} - \text{Actual fixed Overhead}$

Fixed Overhead Expenditure Variance: Fixed Overhead Expenditure Variance is the difference between the Actual Variable Overhead for the period and the budgeted fixed overhead for the period.

$\text{FOEV} = \text{Budgeted fixed overhead} - \text{Actual fixed overhead}$

OR

$(\text{Budgeted hours} \times \text{Std. fixed overhead rate}) - \text{Actual fixed overhead}$

Fixed Overhead Volume Variance: It is the difference between budgeted fixed overhead for the period and the standard fixed overhead for actual production.

$\text{Volume Variance (FOVV)} = (\text{Actual output} \times \text{Standard rate}) - \text{Budgeted fixed overheads}$

OR

$\text{Standard rate} (\text{Actual output} - \text{Standard output})$

OR

$\text{Standard rate per hour} (\text{Standard hours produced} - \text{Budgeted hours})$

OR

$(\text{Absorbed overhead} - \text{Budgeted overhead})$

Fixed Overhead Calendar or Idle Time Variance: It is the difference between Budgeted Fixed overhead for the period and Fixed overhead for days available during the period at standard rate.

FO Calendar Variance = Standard rate (Revised budgeted units – Budgeted units)

OR

Increase or decrease in production due to more or less working days at the rate of revised capacity × Standard rate per unit.

Fixed Overhead Capacity Variance: It is the difference between Budgeted Fixed Overhead for days available at standard rate and Fixed Overhead for actual hours worked at standard rate.

F. O. Capacity Variance = Standard rate (Standard quantity – Budgeted quantity)

Fixed Overhead Revised Capacity Variance (FORCV) = Standard Rate [Standard Quantity – Revised Budgeted Quantity]

Fixed Overhead Efficiency Variance: It is the difference between Fixed Overhead for actual hours worked at standard rate and Standard Fixed Overhead for production.

Fixed Overhead Efficiency Variance (FOEfV) = Standard Fixed Overhead Rate per hour [Standard Production – Actual Production]

STEPS IN CALCULATING FIXED OVERHEAD VARIANCE

FIRST STEP: Basic Computations:

FO1 = Actual Fixed overhead incurred during the period (normally given).

FO2 = Budgeted Fixed Overhead for the period (normally given).

FO3 = Fixed Overheads for the days available at standard rate during the period.

(Budgeted Fixed Overhead for the period/ Budgeted days) × Actual days available

FO4 = Fixed Overhead for actual hours worked at standard rate.

(Budgeted Fixed Overhead for the period / Budgeted hours) * Actual hours

FO5 = Standard Fixed Overhead for production.

(Budgeted Fixed Overhead for the period / Budgeted production) × Actual production

SECOND STEP: Computation of Variances:

Fixed Overhead Expenditure Variance	FO1 – FO2
-------------------------------------	-----------

Fixed Overhead Calendar Variance	FO2 – FO3
Fixed Overhead Capacity Variance	FO3 – FO4
Fixed Overhead Efficiency Variance	FO4 – FO5
Fixed Overhead Variance	FO1 – FO5
Fixed Overhead Volume Variance	FO2 – FO5

THIRD STEP: RECONCILIATION:

Fixed Overhead Variance = Expenditure Variance + Volume Variance

Fixed Overhead Volume Variance = Calendar / Idle Time Variance + Capacity variance + Efficiency Variance

Sales Value Variances



Total Sales Value Variance: It is the difference between Actual Sales Value realised during the period and standard value of sales as per the standard or budget.

SVV = Budgeted Sales – Actual Sales

Sales Value Price Variance: It is the difference between the Actual Sales Value realized for the period and Standard Value of Actual Sales.

SCPV = (Actual Price - Standard Price) x Actual Units Sold

Sales Value Mix Variance: It is the difference between the Standard Value of Actual Sales and the Standard Value of Actual Sales, if these sales had been effected according to the ratio of standard mix.

SVMV = (Actual Volume - Actual Volume at Budgeted Mix) x Budgeted Price

Sales Value Quantity Variance: It is the difference between the Standard Value of Actual Sales, if these sales had been effected according to the ratio of standard mix, and the Standard Value of Sales as per standard or budget.

$$SVQV = (\text{Actual Volume at Budgeted Mix} - \text{Budgeted Volume}) \times \text{Budgeted Price}$$

Sales Value Volume Variance: It is the difference between Standard Value of Actual Sales and the Standard Value of sales as per standard or budget.

$$SVVV = (\text{Actual Units} - \text{Budgeted Units}) \times \text{Standard Price.}$$

STEPS IN CALCULATING SALES VALUE VARIANCE

FIRST STEP: Basic Computations:

SV1 = Actual Sales value realized.

SV2 = Standard Value of Actual Sales.

SV3 = Standard Value of Actual Sales, if it had been in standard proportion.

SV4 = Standard value of sales as per standard or budget.

SECOND STEP: Computation of Variances:

Sales Value Price Variance	SV1 - SV2
Sales Value Mix Variance	SV2 - SV3
Sales Value Quantity Variance	SV3 - SV4
Total Sales Value Variance	SV1 - SV4
Sales Value Volume Variance	SV2 - SV4

THIRD STEP: RECONCILIATION:

Total Sales Value Variance = Sales Value Price Variance + Sales Value Volume Variance

Sales Value Volume Variance = Sales Value Mix Variance + Sales Value Quantity Variance

Benefits of Conducting Variance Analysis:

1. Identify discrepancies

Variance analysis helps identify discrepancies between the actual cash inflows and outflows and the forecasted amounts. By comparing the forecasted cash flow with the actual cash flow, it is easier to identify any discrepancies, enabling the stakeholders to take corrective measures.

2. Refine cash forecasting techniques

Conducting variance analysis allows for a review of past forecasts to identify any errors or biases that may have impacted accuracy. This information can be used to refine forecasting techniques, improve future forecasts, make adjustments to existing forecast templates, or build new ones.

3. Improve financial decision-making

Understanding the reasons for variances can provide valuable insights that can help improve financial decision-making, which is critical in a volatile market. For example, if a variance is caused by unexpected expenses, management may decide to reduce expenses or explore cost-saving measures.

4. Better cash management

By analyzing variances, companies can identify areas where cash management can be improved. This can include better management of accounts receivable or accounts payable, more effective inventory management, or renegotiating payment terms with suppliers.

Accounting Treatment of Variances:

The cost records maintained and entries made under a system of standard costing vary from company to company depending upon the information that is desired from cost records, and the intended use of standard cost and variance analysis. Variances which emerge in standard costing and recorded in the cost books may be disposed of in any of the following ways:

1. Transfer to costing profit and loss account

In this method, the stock of work-in-progress, finished goods and cost of sales are maintained at standard cost and all variances are charged to costing profit and loss account at the end of the accounting period. This method is favoured because standard costs facilities prompt inventory valuation and also variances are separated out so as to attract the attention of the management.

2. Allocation of variances to finished stock, work-in-progress and cost of sales account

Under this method the variances are distributed over stocks of finished goods, work-in-progress and to cost of sales account in proportion to the closing balances (value) of each account depending upon the type of variance.

3. Transfer to reserve account

In this method, favourable variances are carried forward as deferred credits until they are set-off by adverse variances. It is considered that controllable variances are dealt with according to method (ii).

Variance Report

What is a Variance Report?

A variance report is a simple but powerful tool that compares two things: what you expected to happen financially (your budget) and what actually happened (your real numbers). It shows the gaps between the two, called **variances**, so you can see where you're on track and where things went off course.

There's also a more specific version called a price variance report, which zooms in on differences between expected and actual prices paid for goods or services — a must-have for procurement teams keeping an eye on supplier performance and cost control.

Every quality variance report includes a few key parts:

- **Budgeted Amount.** The number you originally planned for, whether it's a cost, a revenue target, or a project budget.
- **Actual Amount.** What was actually spent during the financial reporting period?
- **Variance.** The difference between the budgeted and actual amounts.
- **Variance Percentage Difference.** The size of the variance, shown as a percentage of the budget, makes it easy to spot big swings quickly.

Variance reports tell the story of how your business is operating compared to your expectations. When built and used the right way, they help you take corrective action fast, refine your budgeting process, and improve overall financial performance.

Why Variance Reporting Matters?

Regular variance reporting isn't just a financial formality, but a real-time health check for your business. It plays a critical role in making sure operations are running efficiently and spending stays under control, especially for procurement professionals.

Here's why it matters:

- **Spotting Overspending or Underspending Early.** Variance reports show you quickly where costs are drifting off course. Whether you're spending more than expected or falling short on investments that drive growth, catching these issues early helps you take corrective action before they turn into bigger problems.
- **Enabling Smarter Planning.** When you know exactly where and why things went off plan, you can build better budgets and forecasts going forward. Budget variance insights aren't just backward-looking; they're the key to building more realistic plans.
- **Promoting Financial Transparency.** Clear financial reporting builds trust with executives, board members, and investors. When you can show exactly how budgets are being used, you position your team as responsible, data-driven, and strategically aligned.

Reporting of Variances to Management:

For procurement teams specifically, variance reports highlight supplier performance, help identify maverick spending, and keep strategic sourcing initiatives on track. And for businesses overall, they are the foundation of better financial management, fewer surprises, and more confident decision-making.

The primary purpose of reporting to management is to enable them to take corrective action and arrest unfavourable variances to the extent possible. Therefore, timely and prompt reporting of the variance is of utmost importance. The individual or department responsible for adverse controllable variance should be located. For instance, a variation in the price paid for raw materials would be the responsibility of the purchase manager and a variation in production efficiency is the responsibility of the production manager. The board and the managing director would be concerned with the overall efficiency, with which their plans have been operated by the lower levels of management. The profit and loss account should be prepared in a special manner - starting with the standard or budgeted profit, the various variances would be put in two columns, favourable and unfavorable, and the net results added to or deducted from the standard profit, thus arriving at the actual profit. Management can easily see the factors that have contributed to the change in the profit picture. While reporting the analysis of variances to management, graphs and charts might be used or analysis may be reported in the form of statement and reports giving main details.

In order that variance reporting should be effective, it is essential that the following conditions are fulfilled:

- The variances arising out of each factor should be correctly segregated. If part of a variance due to one factor is wrongly attributed to or merged with that of another, the analysis report submitted to the management would be misleading and wrong inferences may be drawn from it;
- Variances, particularly the controllable variances should be reported with promptness as soon as they occur. This would enable corrective action being taken in time;
- Analysis of uncontrollable variances should be made with the same care as for controllable variances since the analysis of the off standard situation may reveal far reaching effects on the economy of the concern; and
- The forms of reports for the different types of variances should be designed keeping in view the needs of the management and the size of the concern, and no standard forms can, therefore, be suggested.
- It is better to present the profit figures by way of reconciliation of budgeted (or standard) and actual profits on the basis of variances.

When a Variance Report is Used?

Variance reports come into play any time a business needs to double-check if financial expectations match reality. And that's pretty often for small and mid-sized businesses. With tighter budgets and less room for error, staying on top of variances isn't just useful; it's essential.

Here are some common scenarios where businesses can rely on variance reports:

- **Monthly or Quarterly Budget Reviews.** Many SMBs use variance reports at the end of each financial period to review how departments or projects are tracking against their budgets. It's a fast way to flag overspending or unspent funds.
- **Procurement and Vendor Audits.** In procurement, budget variance reports help ensure that purchase prices match what was negotiated. If actual prices consistently come in higher than planned, it's a sign to revisit vendor agreements or sourcing strategies.
- **Project Cost Tracking.** Whether it's a marketing campaign or a product launch, variance reports show if a project is staying within budget. If costs start climbing, project managers can course-correct before the budget gets blown.
- **Cash Flow and Financial Health Monitoring.** For SMBs where cash flow stability is especially critical, tracking budget vs actuals helps forecast available funds more accurately and avoid last-minute financial surprises.

In short, any time a business sets a financial plan and needs to hold itself accountable to that, a variance report becomes an essential tool for managing operations with clarity and confidence.

How to Prepare a Variance Report?

Building a variance report doesn't have to be complicated, but it does require a clear and organized process. When you follow the right steps, you get reports that are accurate, insightful, and easy for stakeholders to act on.

Step 1: Gather Budgeted Data

Start by pulling the approved budget figures for the period you're analyzing. Make sure you have the full breakdown by department, project, or cost category so that you can match actuals against the correct targets. Accuracy at this stage is critical – outdated or incomplete budgets will make your whole report unreliable.

Step 2: Collect Actual Figures

Next, gather actual spending or revenue data from your financial systems, procurement software, or accounting platforms. Double-check that the actuals are final and reconciled. If your data sources aren't consistent, take the time to clean and align them before moving forward.

Step 3: Calculate Variance

Variance Amount = Budgeted Amounts – Actual Amount

Variance Percentage = (Variance Amount/Budgeted Amount) X 100

This gives you both the price difference and the relative size of the variance, making it easier to spot areas that need attention quickly.

Step 4: Analyse the Reason Behind the Variance

A budget variance by itself doesn't tell the full story. You need to understand why it happened. Was it due to supplier price changes, unexpected project delays, volume fluctuations, or internal errors? Digging into root causes gives you the actionable insights needed to correct course or improve future planning.

Step 5: Visualise or Present the Data

Once the calculations and variance analysis are done, package the data clearly. Use tables, charts, or dashboards to present the information in a way that highlights the biggest variances first. Visual reports are much easier for leadership teams to digest and act on compared to raw numbers alone.

Interpreting Variance Report Results

Running the numbers is only half the job. To get real value from a variance report, you need to know how to read the results and when to act on them.

First, understand the basic types of variance:

- **Positive Variance.** When the actual results are better than planned. For example, spending less than budgeted or earning more revenue than expected.
- **Negative Variance.** When actual performance falls short of expectations, such as overspending or missing revenue targets.

Not every variance automatically means there's a problem. Here's what to consider when interpreting your results:

- *Is the Variance Recurring or a One-Time?*

Some variances occur once due to special circumstances (such as a major supplier discount or a short-term price spike). Others repeat month after month, signaling a deeper operational issue that needs fixing.

- *Is It Within Tolerance?*

No budget will ever match actuals perfectly. Set clear thresholds for deviation where minor variances are considered acceptable. Focus your attention on anything outside those limits.

- *Does It Impact Strategic Goals?*

Some deviations may look significant but have little impact in reality. Others, even small ones, might affect critical initiatives like product launches or customer service upgrades.

Thus, the key is to prioritize. Not every variance deserves the same level of attention. Focus your time on variances that are material, recurring, and tied to important business outcomes.

Common Mistakes to Avoid in Variance Reporting

Even the best variance reports can lose their value if a few critical mistakes creep in. Here's what to watch for and how to avoid slipping up.

- **Using Outdated Financial Data**

Old budgets and outdated actuals make financial variance reports meaningless. If you're comparing this quarter's spend to last year's budget, you're not getting an honest view.

Always pull the latest, approved budgets and finalized financial figures. Set clear cutoffs for when data is "report-ready".

- **Not Updating the Procurement Budget Regularly**

Procurement budgets aren't "set and forget". Market conditions change, priorities shift, and new vendors come in. If your budget doesn't reflect these changes, your budget variance report will flag "problems" that aren't actually problems.

Review and update procurement budgets regularly. Ideally, every quarter or when major contracts are renegotiated.

- **Departmental Data Silos**

When procurement, finance, and operations teams don't share information, variance reports end up full of gaps or inconsistencies. Different systems, missing context, and communication breakdowns all add noise to your analysis.

Centralize financial and procurement data where possible, or at least ensure departments are aligned on reporting standards and timelines.

- **Ignoring Small Variances Over Time**

It's easy to dismiss small variances until they snowball into serious budget issues. Tiny overhead costs that repeat month after month can quietly drain cash without raising immediate red flags.

Track even minor variances and look for patterns over time. If small issues are recurring, address them before they grow.

- **Relying Only on Manual Processes**

Manual data entry and spreadsheet-based reporting leave too much room for human error, not to mention slow turnaround times. By the time the report is built, it might already be outdated.

Automate wherever possible. Use procurement and financial platforms that pull live data, auto-calculate variances, and generate real-time dashboards.

From Manual to Automated Variance Reports with Team Procure

Manual budget variance reporting isn't just tedious but risky. Missed entries, outdated numbers, and slow updates can all lead to wrong decisions. **Team Procure** procurement management software gives businesses a smarter way to track budgets versus actuals with:

- **Real-Time [Spend Tracking](#).** Monitor your expenses against budgets in real time. No more waiting for monthly reports to spot issues — get instant visibility and act fast.
- **Automated Data Collection and Custom Reports.** Team Procure automatically pulls in purchasing, invoicing, and budgeting data. It helps you to build variance reports without manual input, reducing errors and saving hours of work.
- **Budget Thresholds and Automated Alerts.** Set custom budget limits by project, department, or vendor. Get instant alerts when you're nearing a threshold, so you can address problems before they spiral.
- **Seamless Integration with Accounting Software.** Sync easily with platforms like QuickBooks. Keep your financial records aligned without double data entry or manual uploads.

By simplifying the price variance reporting process, Team Procure helps procurement teams, finance leaders, and SMB owners stay on top of their budgets with far less effort — and with a lot more confidence.

Sample of Variance Report:

FISCAL YEAR 2017-2018 BUDGET vs ACTUAL VARIANCE ANALYSIS QUARTER ENDING DECEMBER 31, 2017

Account Number	Account Name	Dec 2017 Financial Report Page #	Original Budget	Revised Budget	Actual to December 31, 2017	% of Budget	Variance Explanation	Recommended Action
GENERAL FUND REVENUES								
001.0000.311.010000	Ad Valorem Taxes	13	6,428,767	6,428,767	5,336,828	83.0%	Ad valorem tax revenues typically received in November to December due to timing of tax bills and discounts for early payment.	Variance within normal parameters
001.0000.312.041000	Local Option Gas Tax	13	488,436	488,436	44,245	9.1%	State distributed tax revenues received during October and November 2017 are accrued back as revenues for the previous fiscal year. Only one month of revenue (collected on December 2017) has been recognized YTD.	Variance within normal parameters
001.0000.315.000100	Local Communication Serv Tax	13	1,186,028	1,186,028	116,396	9.8%	State distributed tax revenues received during October and November 2017 are accrued back as revenues for the previous fiscal year. Only one month of revenue (collected on December 2017) has been recognized YTD.	Variance within normal parameters
001.0000.335.018000	Half Cent Sales Tax	13	1,195,056	1,195,056	91,468	7.7%	State distributed tax revenues received during October and November 2017 are accrued back as revenues for the previous fiscal year. Only one month of revenue (collected on December 2017) has been recognized YTD.	Variance within normal parameters
001.0000.347.000104	Recreation Department Income	13	330,000	330,000	39,733	12.0%	Recreation Department income is seasonal and increases during summer months.	Variance within normal parameters
001.0000.354.000100	Fines & Forfeitures	13	75,000	75,000	32,470	43.3%	Driven primarily by an increase in parking fines during the first quarter of FY 2017.	None required.
001.0000.361.010200	Interest on Investments	13	100,000	100,000	(16,210)	-16.2%	The City's short term investments experienced first quarter losses in value.	Continue to monitor.
001.0000.362.000114	Recreation Facilities Rental	13	140,000	140,000	20,335	14.5%	Recreation Facilities Rental income is seasonal and increases during spring months.	Variance within normal parameters
GENERAL FUND EXPENDITURES								
122002	Contrib to Police Pension	18	779,802	779,802	779,802	100.0%	Per actuarial valuation, entire 2017-2018 police pension contribution was made in October 2017, in order to reduce required contribution due to timing.	Variance within normal parameters
122003	Contribution to Fire Pension Fund	18	72,765	72,765	-	0.0%	City contributions to be made after receiving final State contributions (anticipated in September 2018)	Variance within normal parameters
123006	Contribution to OPEB Trust	18	452,349	452,349	235,510	52.1%	True-up of OPEB Trust reimbursements to be made in second quarter of FY 2018.	Variance within normal parameters
331001	Professional Services	18	125,803	126,402	7,814	6.2%	Annual expenditures which vary according to timing of services rendered and are not expended on a prorata basis.	None required.
345001	General Insurance	19	390,870	390,870	190,837	48.8%	Insurance payments are made on a quarterly basis and are due at the beginning of each quarter. As of December 31st, 2017 the City has made two quarterly insurance payments.	Variance within normal parameters
346003	Building Maintenance	19	86,750	86,750	12,581	14.5%	Expenses are variable based on actual maintenance requirements. Not expended on a straight pro-rata basis.	None required.
346028	Main Traffic Signals	19	85,000	85,000	2,260	2.7%	Expenses are variable based on actual signal maintenance requirements. Not expended on a straight pro-rata basis.	None required.
346200	Software Maintenance	19	189,757	186,553	87,784	47.1%	Annual expenditures which vary according to renewal terms of existing software maintenance agreements.	Variance within normal parameters
349200	Hurricane Irma Expenses	19	-	-	181,754	0.0%	Storm related expenditures, primarily debris removal and monitoring (\$157.0K) and beach sand (\$16.0K), not included in original budget.	Continue to monitor. Anticipate FEMA reimbursement to offset expenditures.
352007	Sign Material	19	70,250	70,250	33,631	47.9%	Based on a projection of the years' workload, the sign shop buys the material by truckload at the beginning of the year to save on both material and shipping cost. Material orders during the first quarter amount to \$33.6K.	None required.
359900	Interfund Services	20	(225,000)	(225,000)	(86,988)	38.7%	Interfund services charges vary according to actual services rendered and are not expended on a prorata basis.	None required.
605002	Portable Radios	21	79,906	79,906	-	0.0%	This budget line item is for a one time capital purchase of a multi-year radio upgrade. FY17-18 increment to be ordered later this year.	None required.

PROBLEMS ON STANDARD COSTING

1. The standard material inputs required for 1,000 kg of a finished product are given below:

Material	Quantity in Kg	Standard Rate per Kg in Rs.
P	450	20
Q	400	40
R	250	60
Total	1,100	
Standard Loss	100	
Standard output	1,000	

Actual production in a period was 20,000 Kg. of the finished product for which the actual quantities of material used and the prices paid thereof are as under:

Material	Quantity in Kg	Purchase Price per Kg in Rs.
P	10,000	19
Q	8,500	42
R	4,500	65

Calculate the:

- a) Material Cost Variance b) Material Price Variance c) Material Usage Variance
- d) Material Mix Variance e) Material Yield Variance
2. XYZ Company manufactures a product ABC by mixing three raw materials. For every 100 Kg of ABC, 125 Kg. of materials are used. In April 2017, there was an output of 5,600 kg. of ABC. The Standard and actual particulars of April 2017 are as under:

Raw Material	Standard		Actual	
	Mix in %	Price per Kg. in Rs.	Mix in %	Price per Kg. in Rs.
X	50	40	60	42
Y	30	20	20	16
Z	20	10	20	12

Calculate all the Material Variances.

3. The standard cost of a certain chemical mixture is as under:
 40% of Material A at Rs. 20 per tonne. 60% of Material B at Rs. 30 per tonne.
 A standard loss of 10% is expected in production. The following actual cost data is given for the period.
 180 tonnes material A at a cost of Rs. 18 per tonne.
 220 tonnes material B at a cost of Rs. 34 per tonne.

The weight produced is 364 tonne.

Calculate the:

b) Material Cost Variance b) Material Price Variance c) Material Usage Variance

d) Material Mix Variance e) Material Yield Variance

4. The following information has been extracted from the records of a chemical company:

Standard Price : Raw Material A = Rs. 2 per Kg. Raw Material B = Rs. 10 per Kg.

Standard Mix : A = 75% and B = 25% (by weight)

Standard yield = 90%

In a period of actual costs, usages and output were as follows:

Used:

2200 Kgs. Of A costing Rs. 4,650.

800 Kgs. Of B costing Rs. 7,850.

Output: 2850 Kgs of products.

Calculate Material Cost Variances.

Also Reconcile the Material Cost Variances.

5. From the following data compute the Material Cost Variances and Reconcile the Material Variances.

Materials	Standard		Actual	
	Units	Price per unit	Units	Price per unit
Zee	3500	10	3700	12
Wee	1500	21	1650	20
Tee	1000	33	1250	36

6. The standard mix of product A2 is as follows:

Kgs.	Materials	Price per Kg. (Rs.)
45	X	6
25	Y	4.50
30	Z	9.50

The Standard loss in production is 10% of input. There is no scrap value.

Actual production for a month was 7,425 Kgs. Of A2 from 80 mixes. Actual consumption and purchases of material during the month were:

Kgs.	Materials	Price per Kg. (Rs.)
4,200	X	6.50
1,700	Y	4.25
2,600	Z	9.75

You are required to calculate the variances Material Variances and reconcile the same.

7. The standard cost for producing 180 Kg of a product whose raw material inputs are A and B is given below:

Material A - 60 Kg at Rs. 10 per Kg	Rs. 600
Material B - 140 Kg at Rs. 2 per Kg	Rs. 280
Total Cost	Rs. 880

The actual prices of A and B were Rs. 12 and Rs. 8 per Kg respectively. Consumption of B was 108 Kg. The actual output at 80% yield was 144 Kg. Calculate all the Material Variances and reconcile them.

8. Following information is given regarding standard composition and standard rates of a gang of workers:

Standard Composition	Standard hourly rate
10 Men	Re. 0.625
5 Women	Re. 0.400
5 Boys	Re. 0.350

According to given specifications, a week consists of 40 hours and standard output for a week is 1,000 units.

In a particular week, gang consisted of 13 men, 4 women and 3 boys and actual wages were paid as follows:

Men at Re. 0.6 per hour

Women at Re. 0.425 per hour

Boys at Re. 0.325 per hour

Two hours were lost in the week due to abnormal idle time. Actual production was 960 units in the week.

Find out:

- Labour Rate Variance
- Labour Mix Variance
- Labour Idle Time Variance
- Labour Yield Variance
- Labour Efficiency Variance
- Labour Cost Variance.

You are also required to reconcile the above variances.

9. The following was the composition of a gang of workers in a factory during a particular month, in one of the production departments. The standard composition of workers and wage rate per hour were as follows:

Skilled	Two workers at a standard rate of Rs. 20 per hour each.
Semi-skilled	Four workers at a standard rate of Rs. 12 per hour each.

Unskilled	Four workers at a standard rate of Rs. 8 per hour each.
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The standard output of the gang was four units per hour of the product.

During the month in question, however, the actual composition of the gang and hourly rates paid were as under:

Nature of worker	No. of workers	Wage rate paid per worker per hour engaged (Rs.)
Skilled	2	20
Semi-skilled	3	14
Unskilled	5	10

The gang was engaged for 200 hours during the month, which included 12 hours when no production was possible, due to machine break-down. 810 units of the product were recorded as output of the gang during the month.

You are required:

- g) Labour Rate Variance
- h) Labour Mix Variance
- i) Labour Idle Time Variance
- j) Labour Yield Variance
- k) Labour Efficiency Variance
- l) Labour Cost Variance.

You are also required to reconcile the above variances.

10. The standard labour complement and the actual labour complement engaged during the month are given below:

Particulars	Skilled	Semi-skilled	Unskilled
Standard number of workers in a group	30	10	10
Standard wage rate per hour (Rs.)	5	3	2
Actual number of worker employed during the month in the group	24	15	12
Actual wage rate per hour (Rs.)	6	2.5	2

During the month of 200 working hours (which included 15 hours when no production was possible due to machine break-down), the group produced 9,600 units. Standard output of the gang was 50 units per hour.

11. Following information is obtained from M/S Will and Urvish Co. Ltd.

Budgeted production for the period	600 units
Budgeted variable overhead	Rs. 15,600
Standard time for one unit	20 hours
Actual production for the period	500 units
Actual variable overhead	Rs. 14,000

Actual hours worked	9,000
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Calculate the following variances:

- Variable overhead expenditure variance
- Variable overhead efficiency variance
- Variable overhead variance

You are also required to reconcile the same.

12. From the following information pertaining to January, calculate the Overhead Variances:

Particulars	Production units	Variable Overheads (Rs.)	Hours Worked
Budgeted	300	7800	(See Note 1)
Actual	250	7000	4500 (See Note 2)

Note: 1: Standard time for 1 unit is 20 hours.

Note: 2: This includes 300 hours of abnormal idle time.

13. The following is data is given. Calculate Variable Overhead Variances.

Particulars	Production units	Man hour to produce the output	Variable Overheads in (Rs.)
Budgeted	400	8,000	10,000
Actual	360	7,000	9,150

14. From the following cost data, calculate the Fixed Overhead Variances and reconcile the same.

	Budgeted	Actual
No. of working days	20	22
Man-hours per days	8000	8400
Output per man-hour in units	1.0	.9
Overhead Cost (Rs.)	160000	168000

15. Cost data given for Nishu & Co. for July 1993 is as follows:

	Budgeted	Actual
Fixed Overheads	Rs. 10,000	Rs. 10,200
Units of production	5,000	5,200
Standard time for one unit	4 hours	-
Actual hours worked	-	20,100 hours

Find out fixed overhead variances and reconcile them.

16. From the following data, calculate:

- a) Sales Value Variance
- b) Sales Price Variance
- c) Sales Mix Variance
- d) Sales Quantity Variance
- e) Sales Value Volume Variance

	Qty (kg)	Std. sales price Rs. Per kg.	Total Cost (Rs.)	Qty (kg)	Actual sales price Rs. Per kg.	Total Cost (Rs.)
Product X	500	5	2500	500	5	2500
Product Y	400	6	2400	600	6.25	3750
Product Z	300	7	2100	400	6.75	2700
	1200		7000	1500		8950

17. The following information of Lucky & Co. is given for December 207.

Budgeted Sales:

Product A = 500 units @ Rs. 2 per unit (Standard Cost Rs. 1.75 per unit)

Product B = 700 units @ Rs. 1.50 per unit (Standard Cost Rs. 1.30 per unit)

Actual Sales:

Product A = 560 units @ Rs. 1.95 per unit.

Product B = 710 units @ Rs. 1.40 per unit.

Find out:

- a) Sales Margin Price Variance
- b) Sales Margin Mix Variance
- c) Sales Margin Quantity Variance
- d) Total Sales Margin Variance
- e) Sales Margin Volume Variance
